US 20250271295A1

(19) United States
(12) Patent Application Publication
Suh et al.
(10) Pub. No.: US 2025/0271295 A1
(43) Pub. Date: Aug. 28, 2025

(54) SELF-POWERED OPTICAL AMPLIFIERS

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(21) Appl. No.: **18/587,306**

(22) Filed: **Feb. 26, 2024**

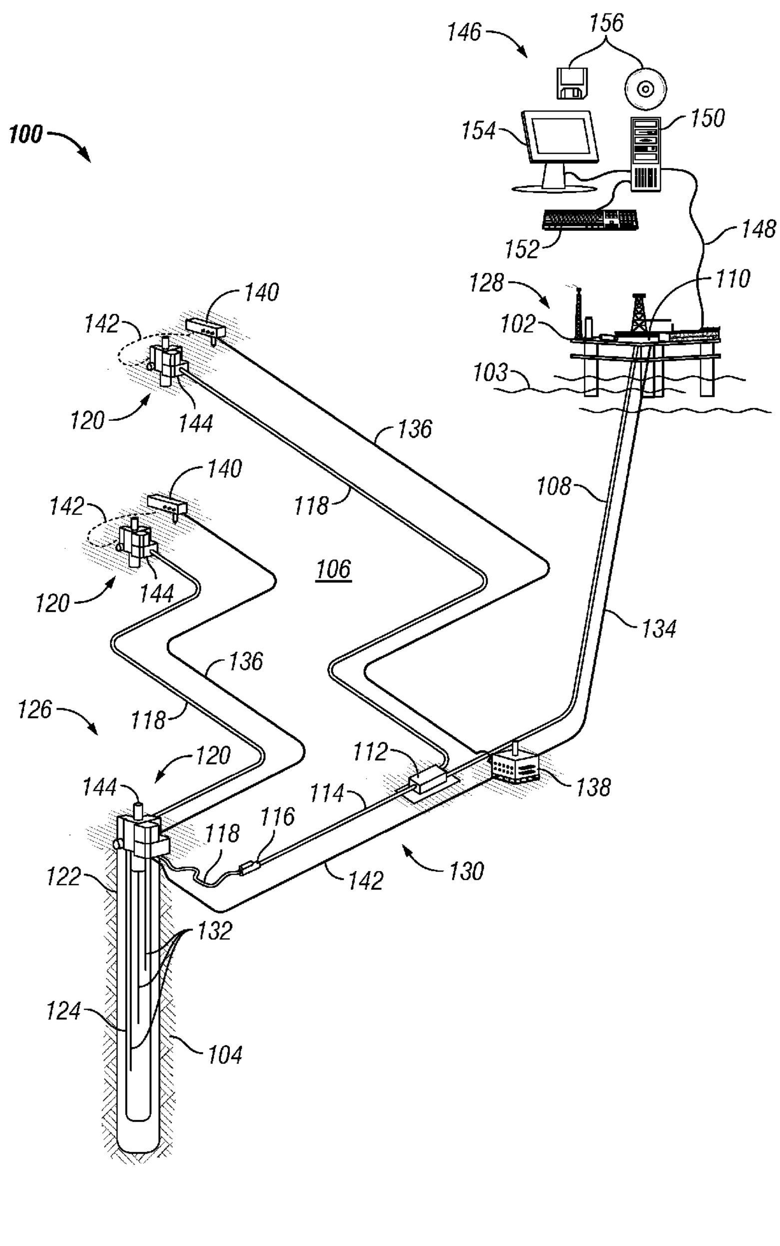
Publication Classification

(51) Int. Cl.
G01H 9/00 (2006.01)
G01V 1/22 (2006.01)

(52) U.S. Cl.
CPC *G01H 9/004* (2013.01); *G01V 1/226* (2013.01)

(57) **ABSTRACT**

A self-powered optical amplifier (SPOA) may include a proximal tap connected to a fiber optic cable, a proximal wavelength division multiplexer (WDM) optically connected to the proximal tap, and an Erbium doped fiber optically connecting the proximal WDM to a distal WDM. The SPOA may further includes a distal tap optically connected to the distal WDM, a pump optically connected to the distal WDM, and a controller optically connected to the proximal tap, the distal tap, and further electrically connected to the pump and a power supply.



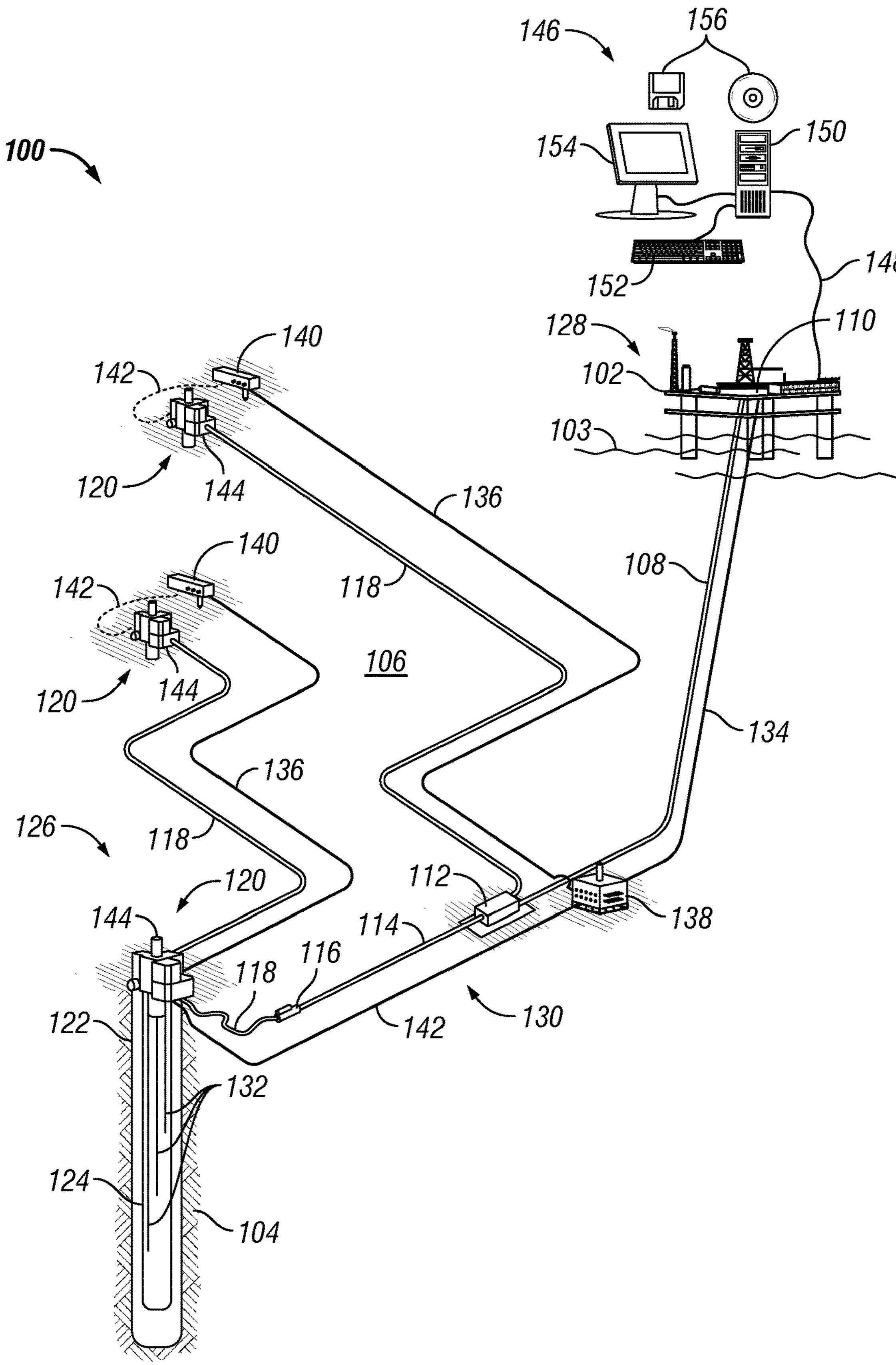

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FIG. 1A

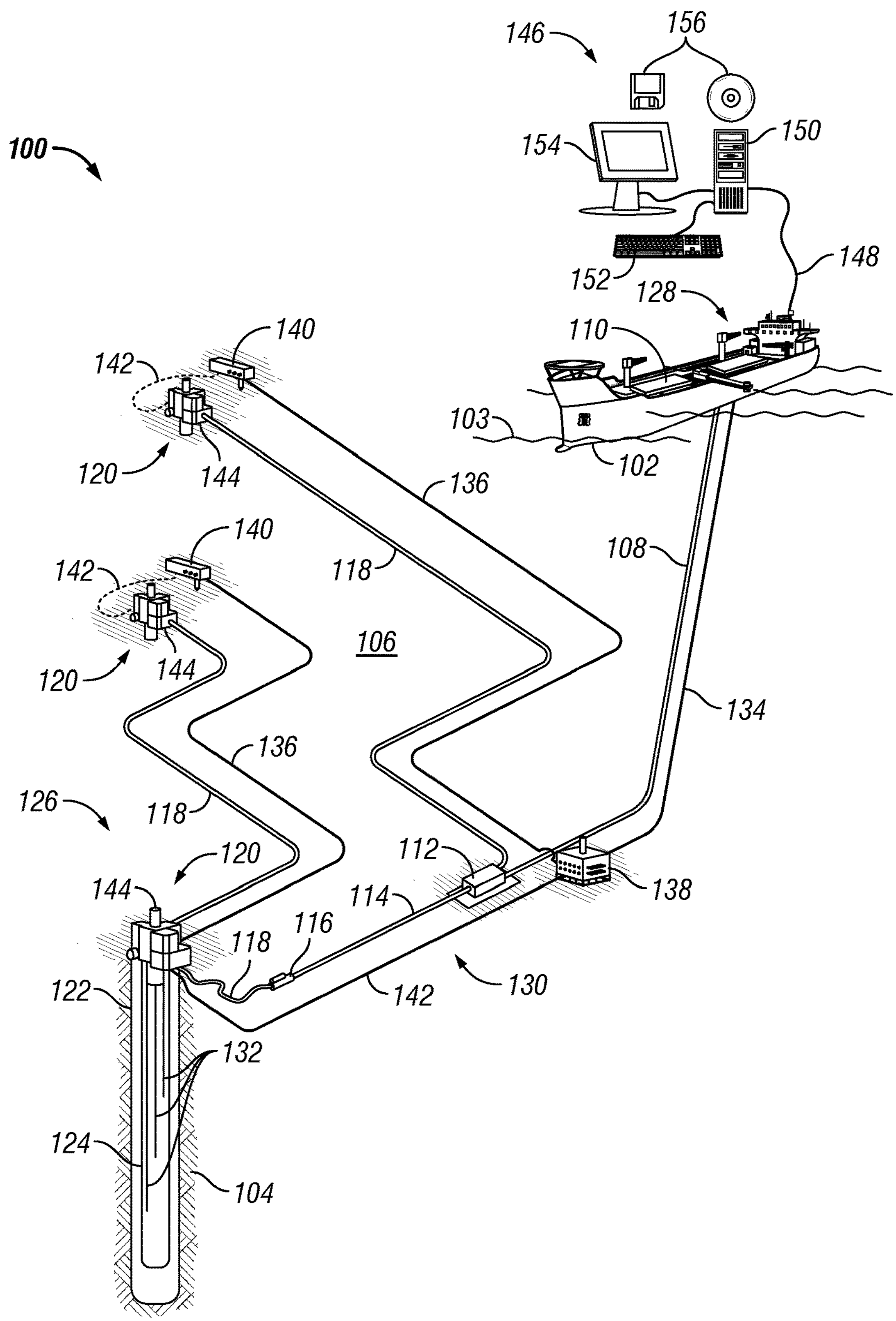
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FIG. 1B

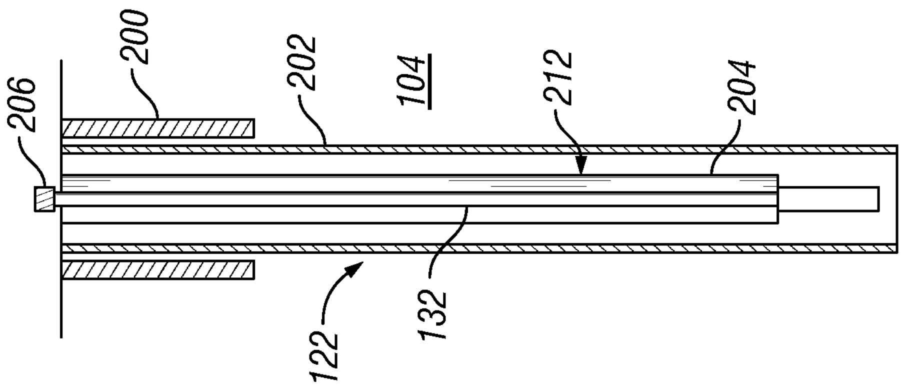


FIG. 2C

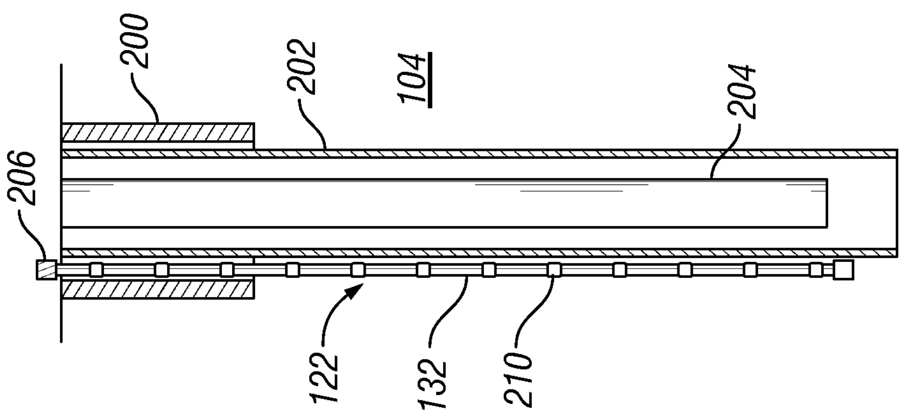


FIG. 2B

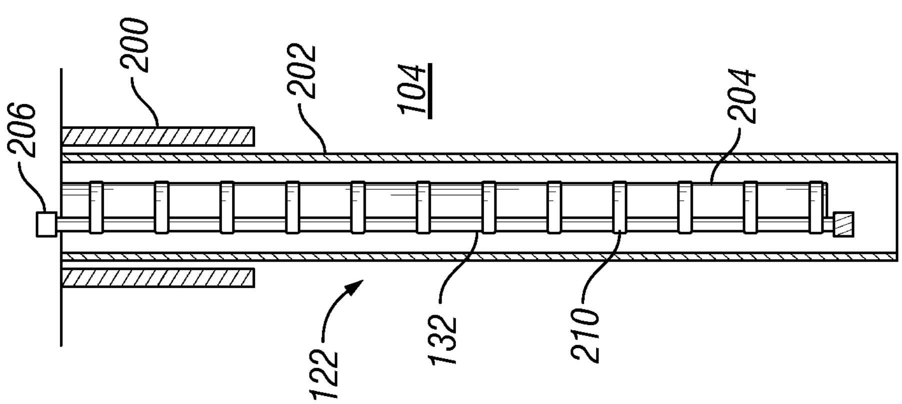


FIG. 2A

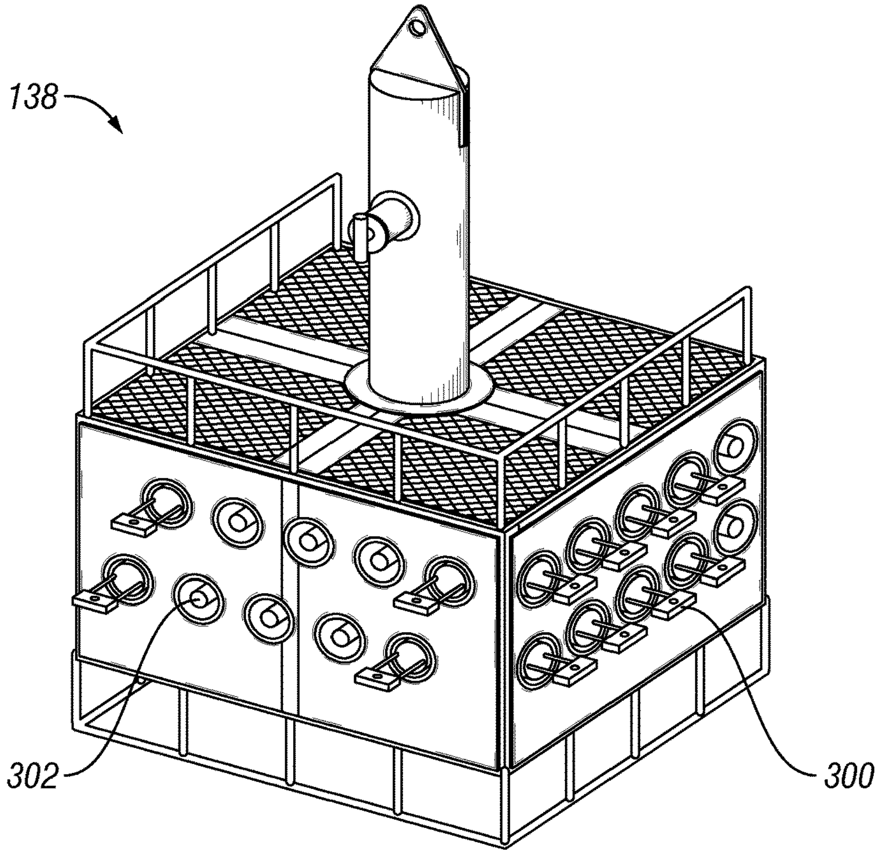


FIG. 3

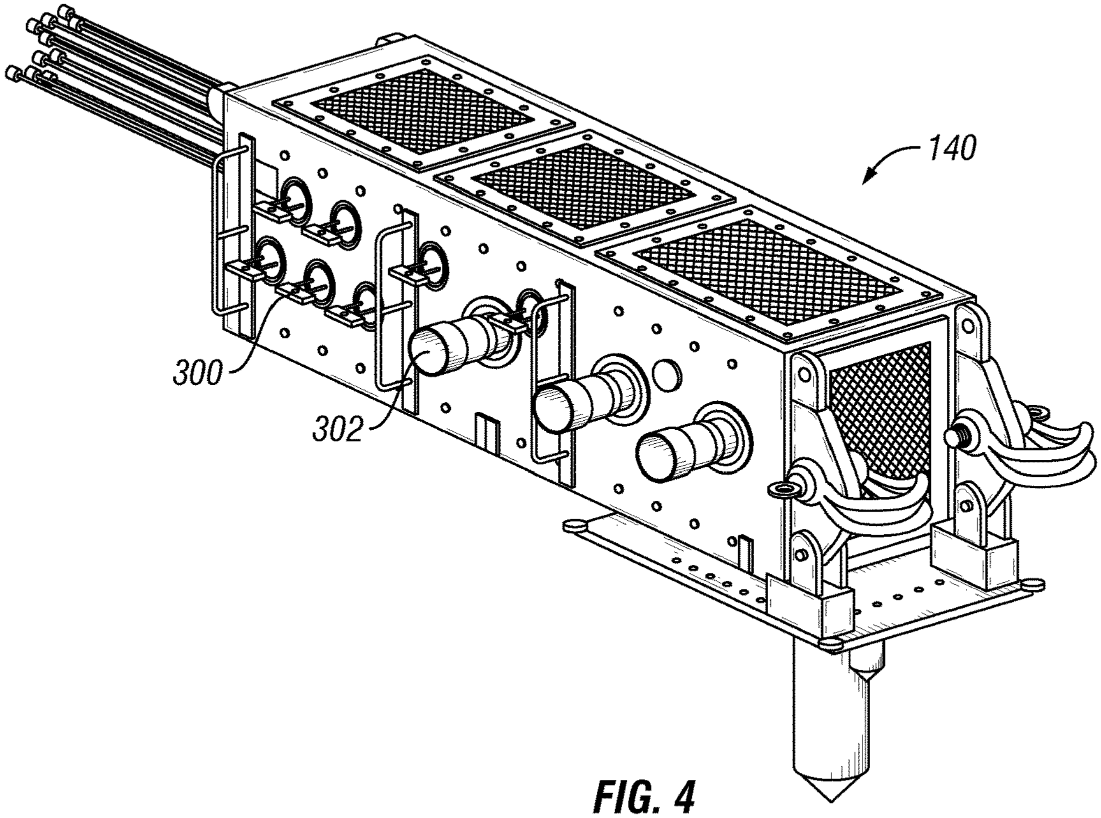
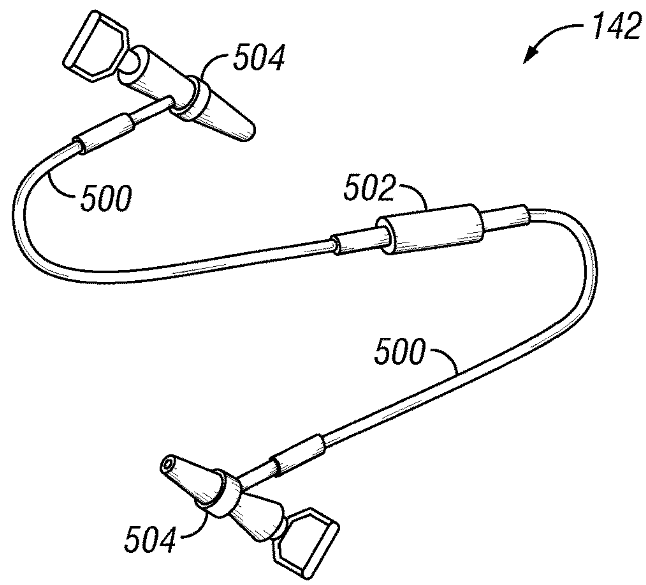


FIG. 4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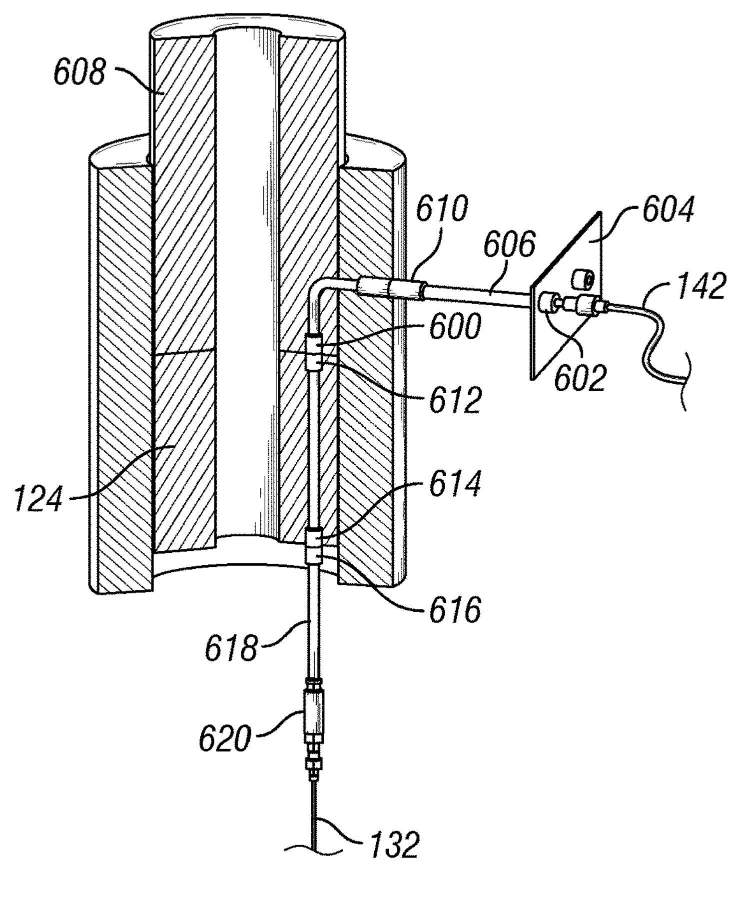
502
500
504

FIG. 5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618
620
132

FIG. 6B

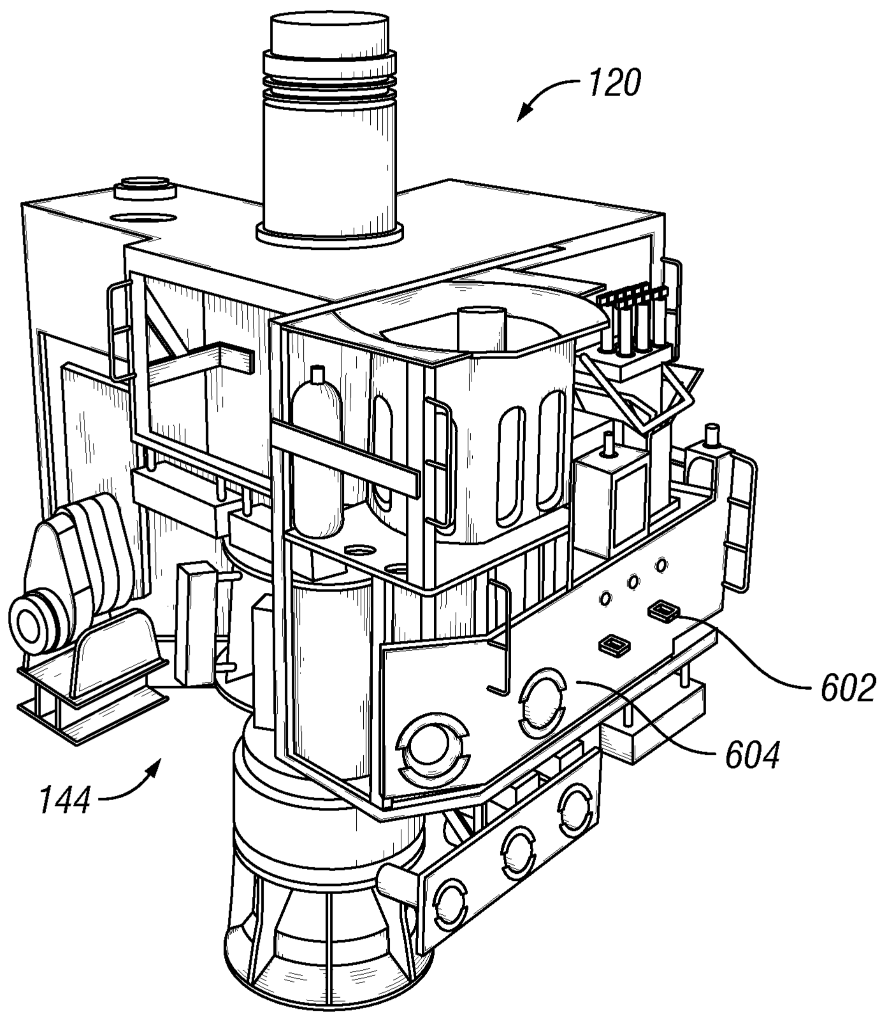
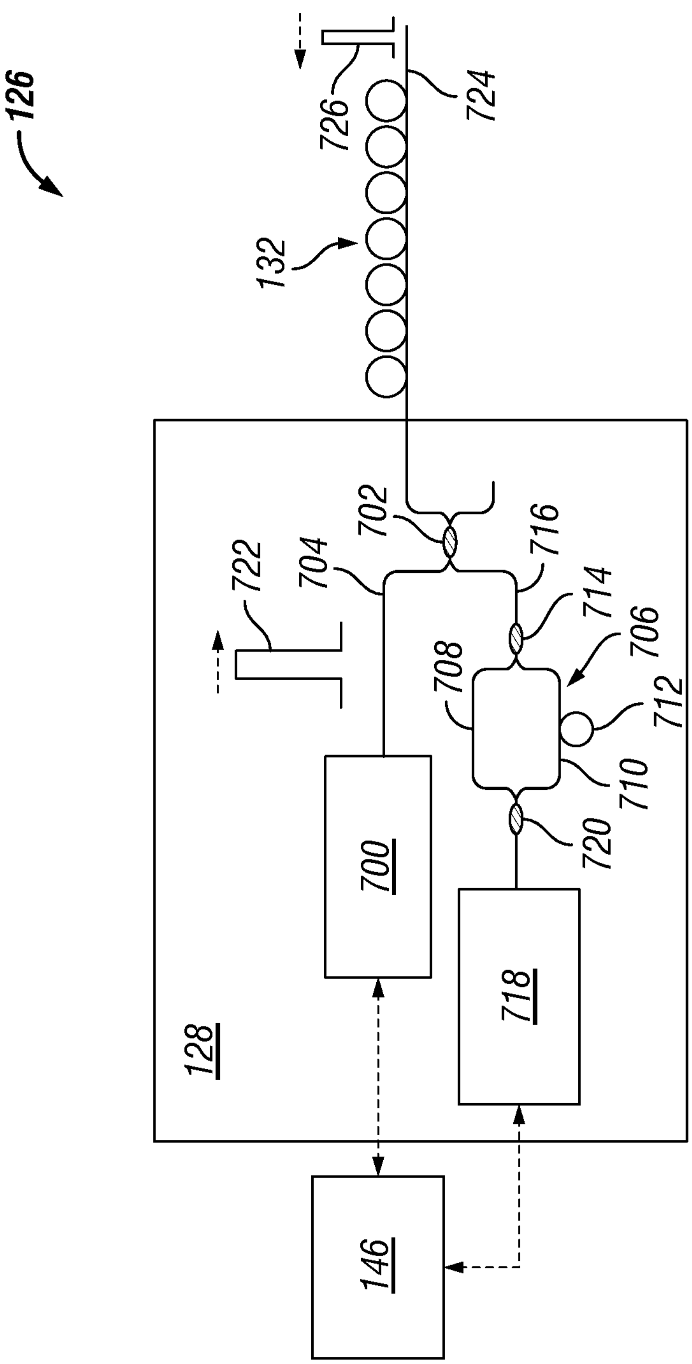


FIG. 6A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726
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702
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712
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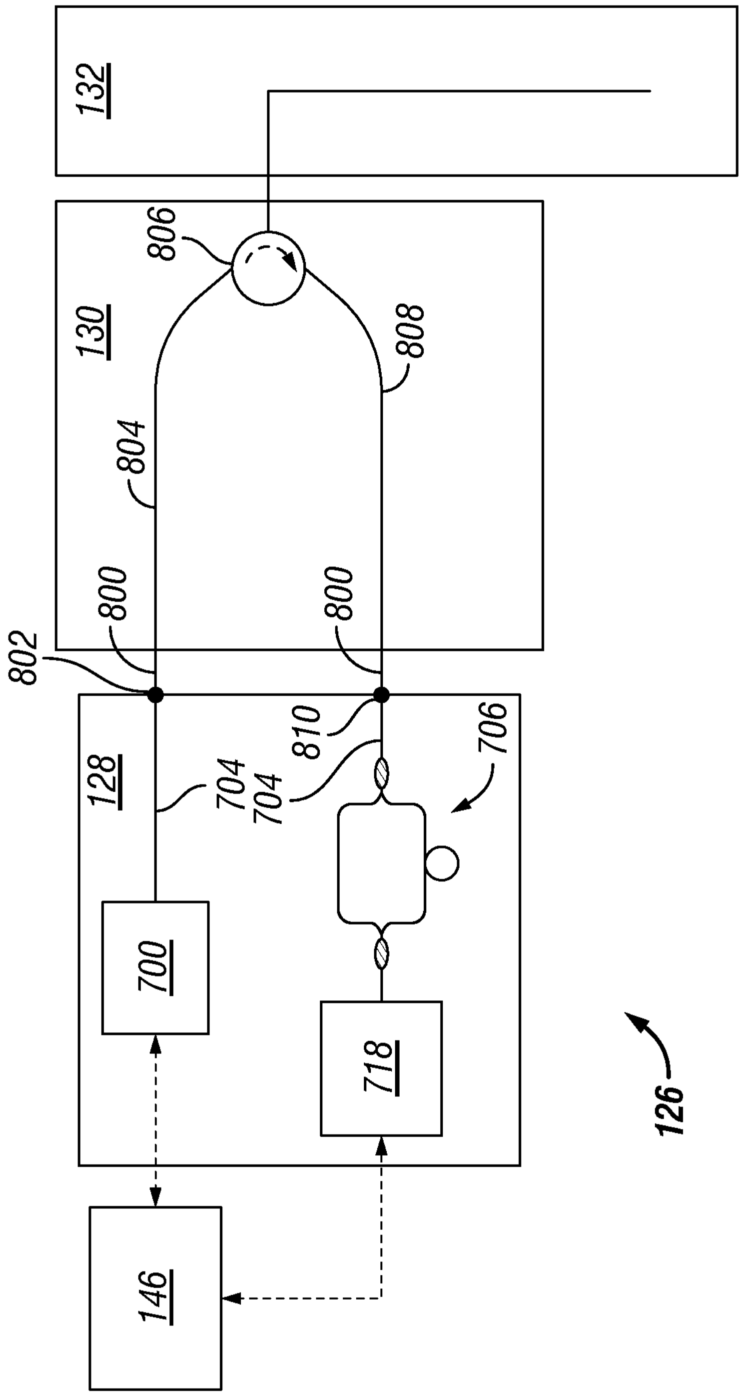
128
146

FIG. 7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146
126

FIG. 8

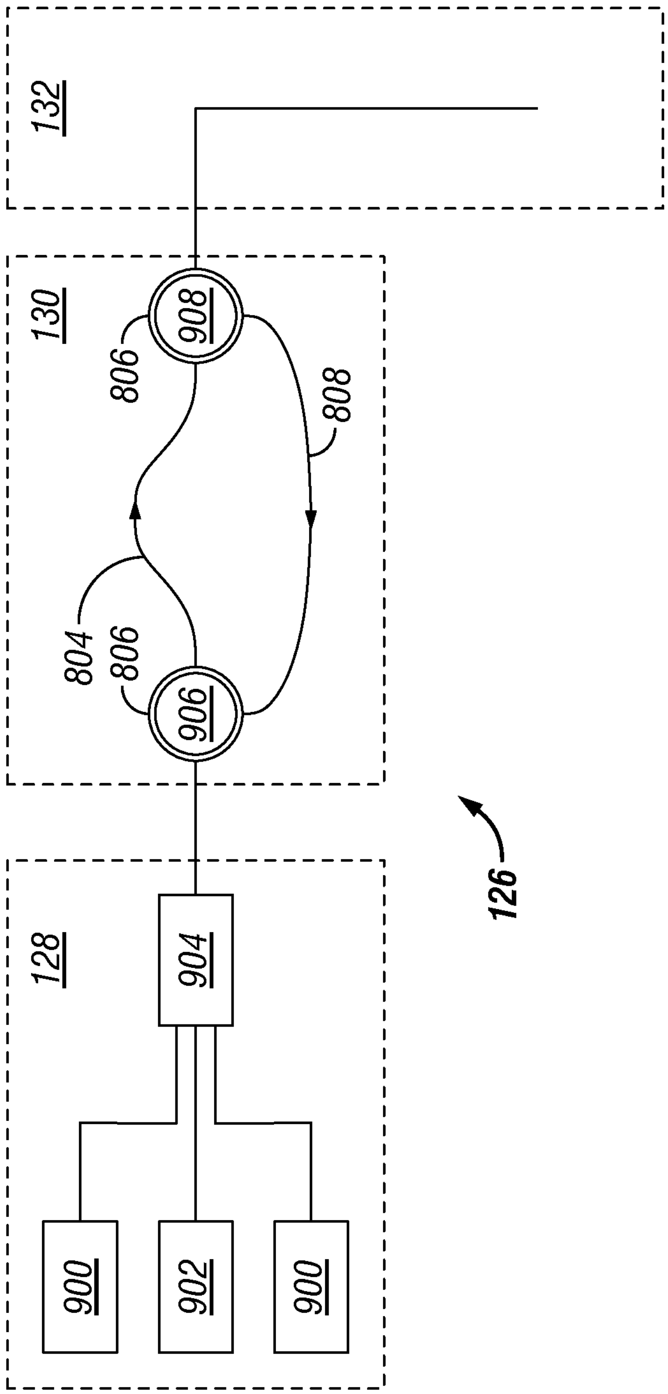


FIG. 9

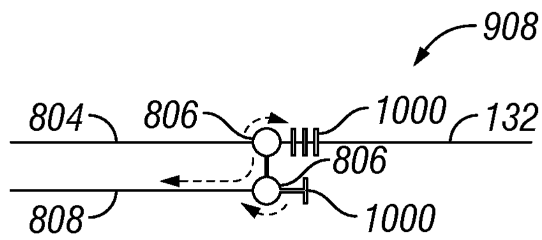


FIG. 10

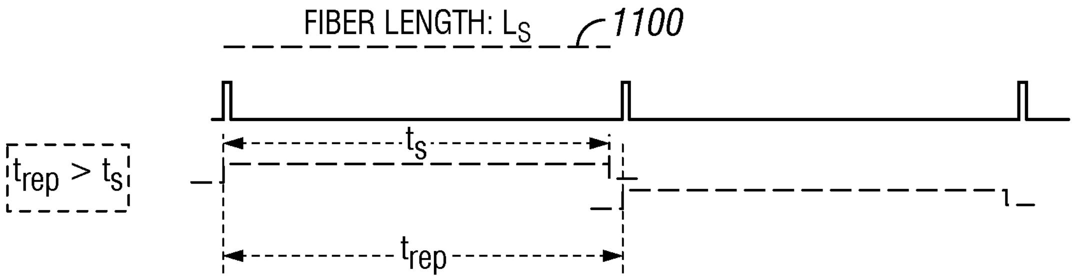


FIG. 11

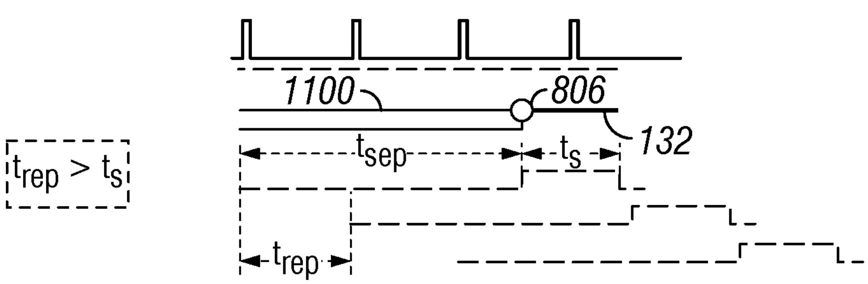


FIG. 12

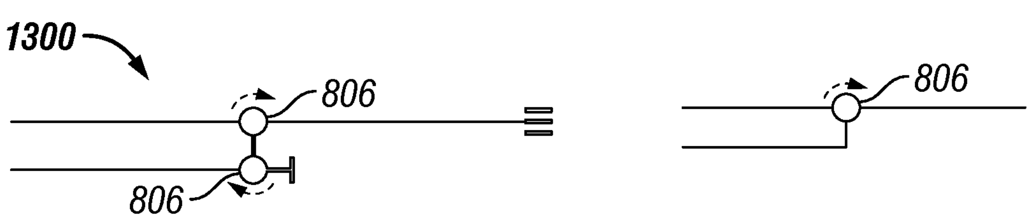


FIG. 13

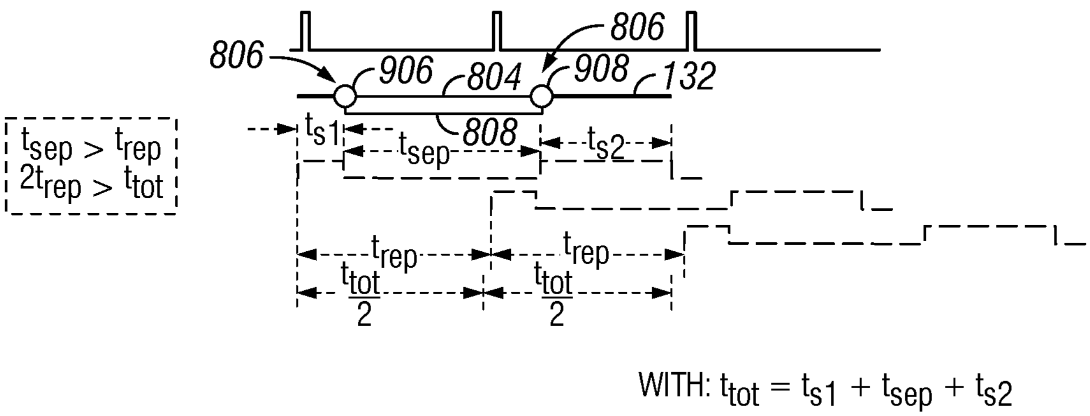


FIG. 14

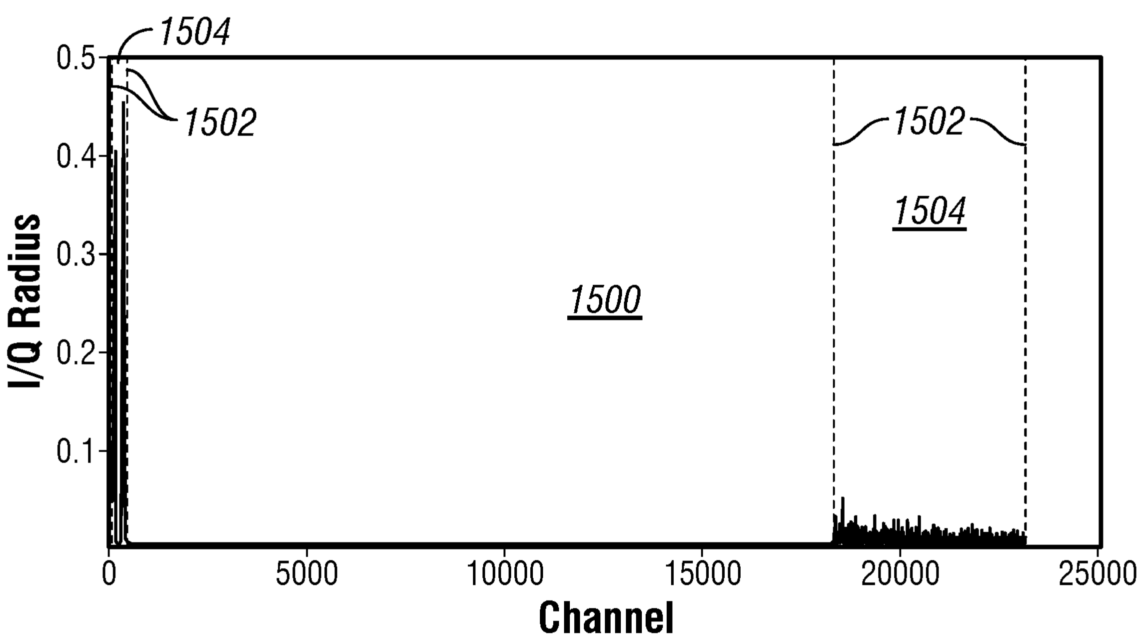


FIG. 15A

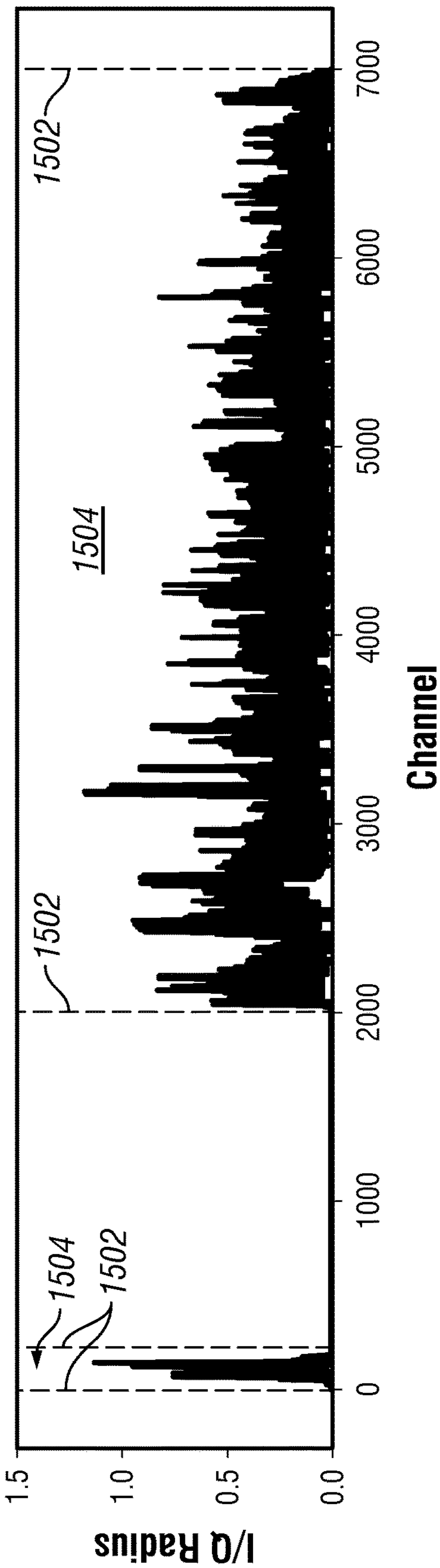


FIG. 15B

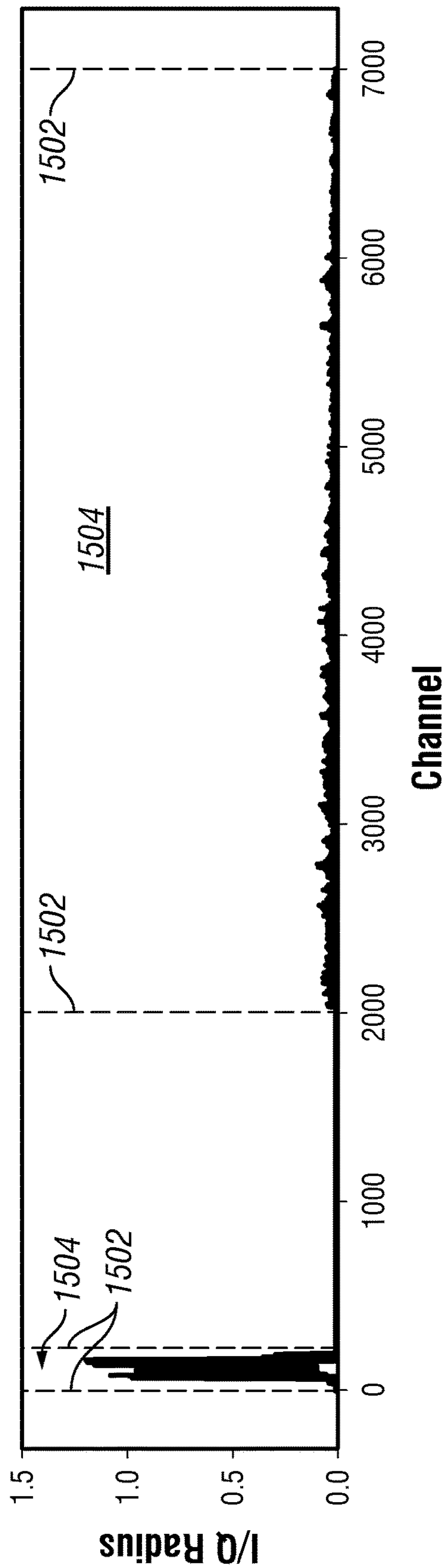


FIG. 15C

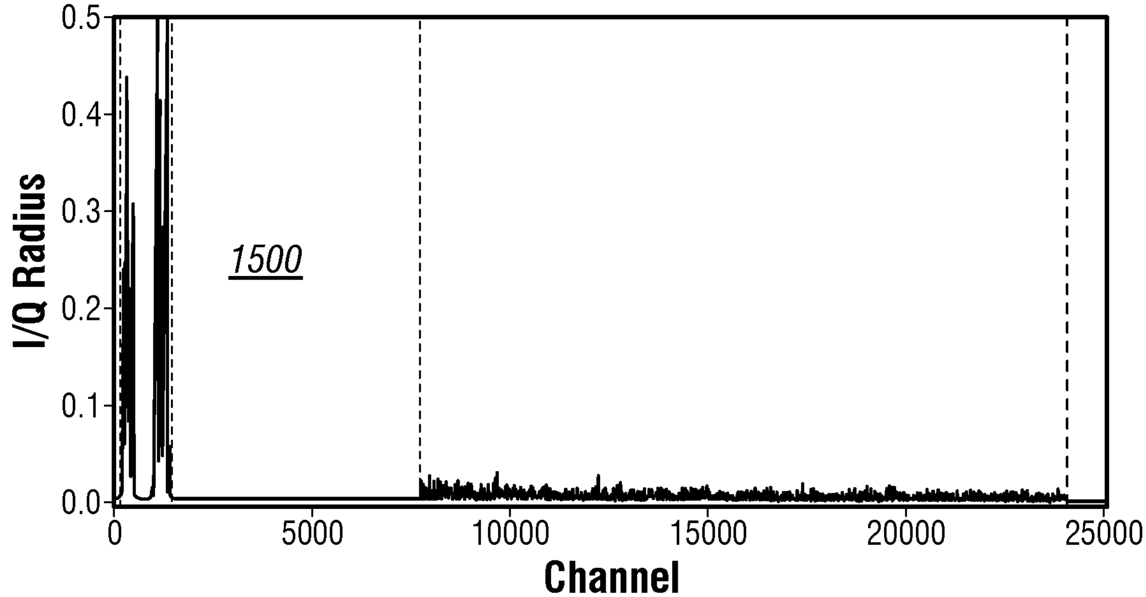

0.5
0.4
0.3
0.2
0.1
0.0
I/Q Radius
1500
0
5000
10000
15000
20000
25000
Channel

FIG. 16

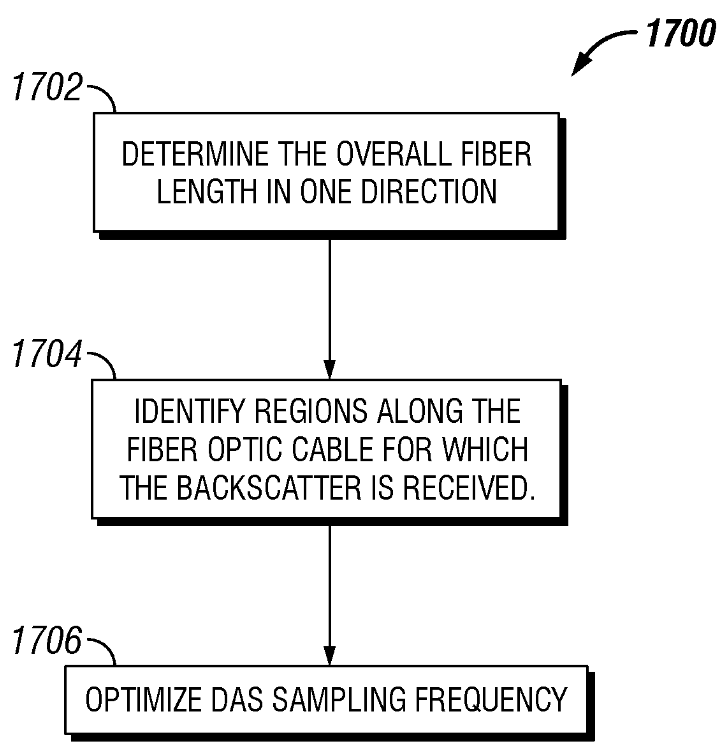

1700
1702
DETERMINE THE OVERALL FIBER LENGTH IN ONE DIRECTION
1704
IDENTIFY REGIONS ALONG THE FIBER OPTIC CABLE FOR WHICH THE BACKSCATTER IS RECEIVED.
1706
OPTIMIZE DAS SAMPLING FREQUENCY

FIG. 17

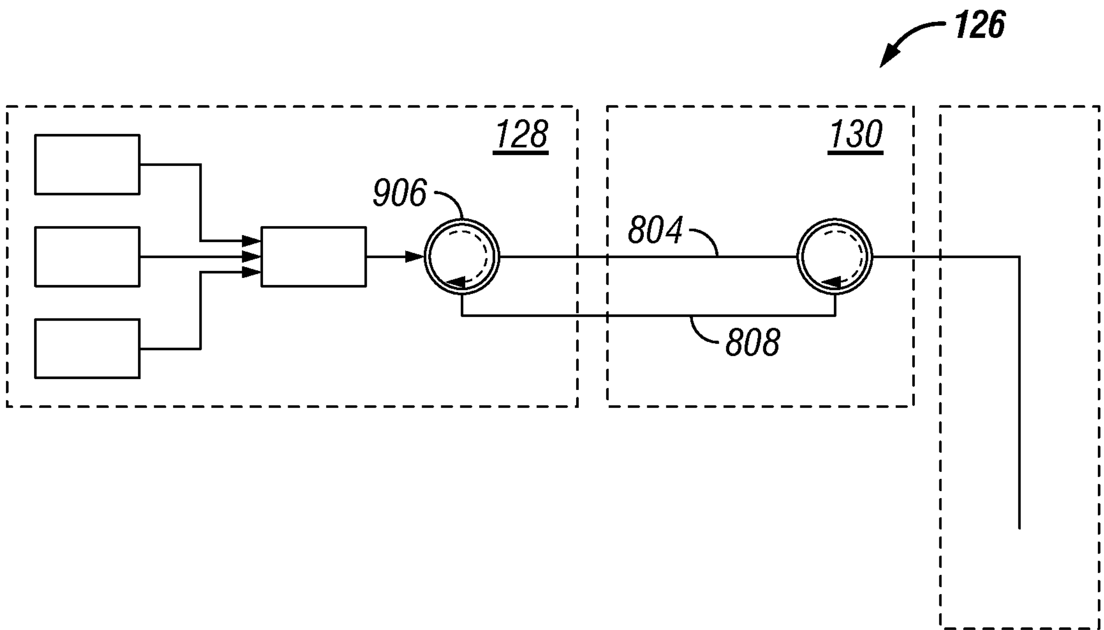


FIG. 18

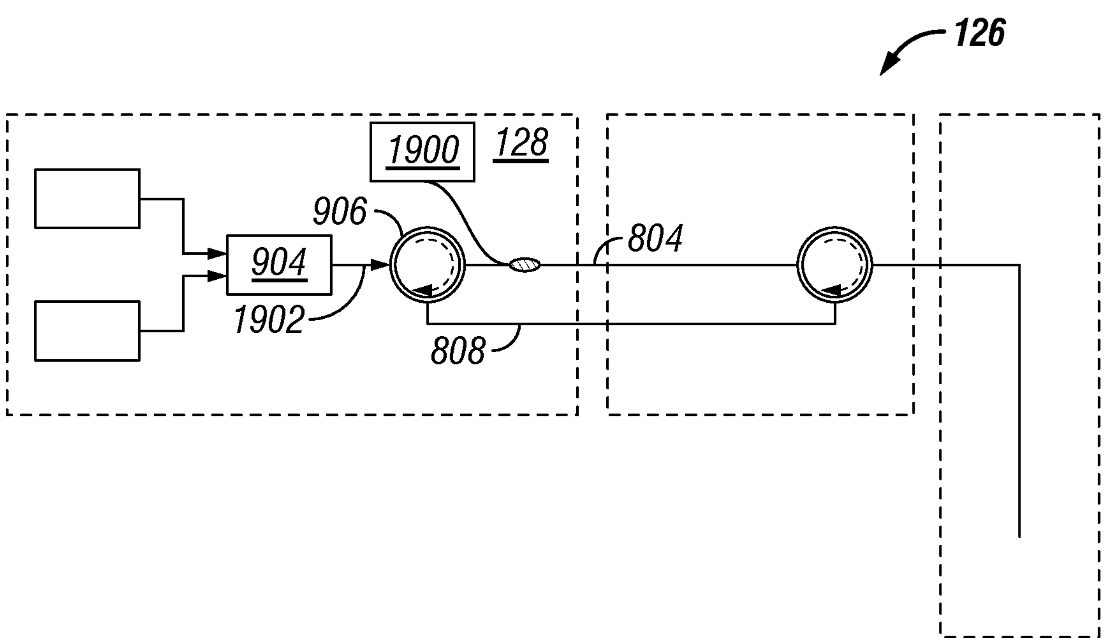


FIG. 19

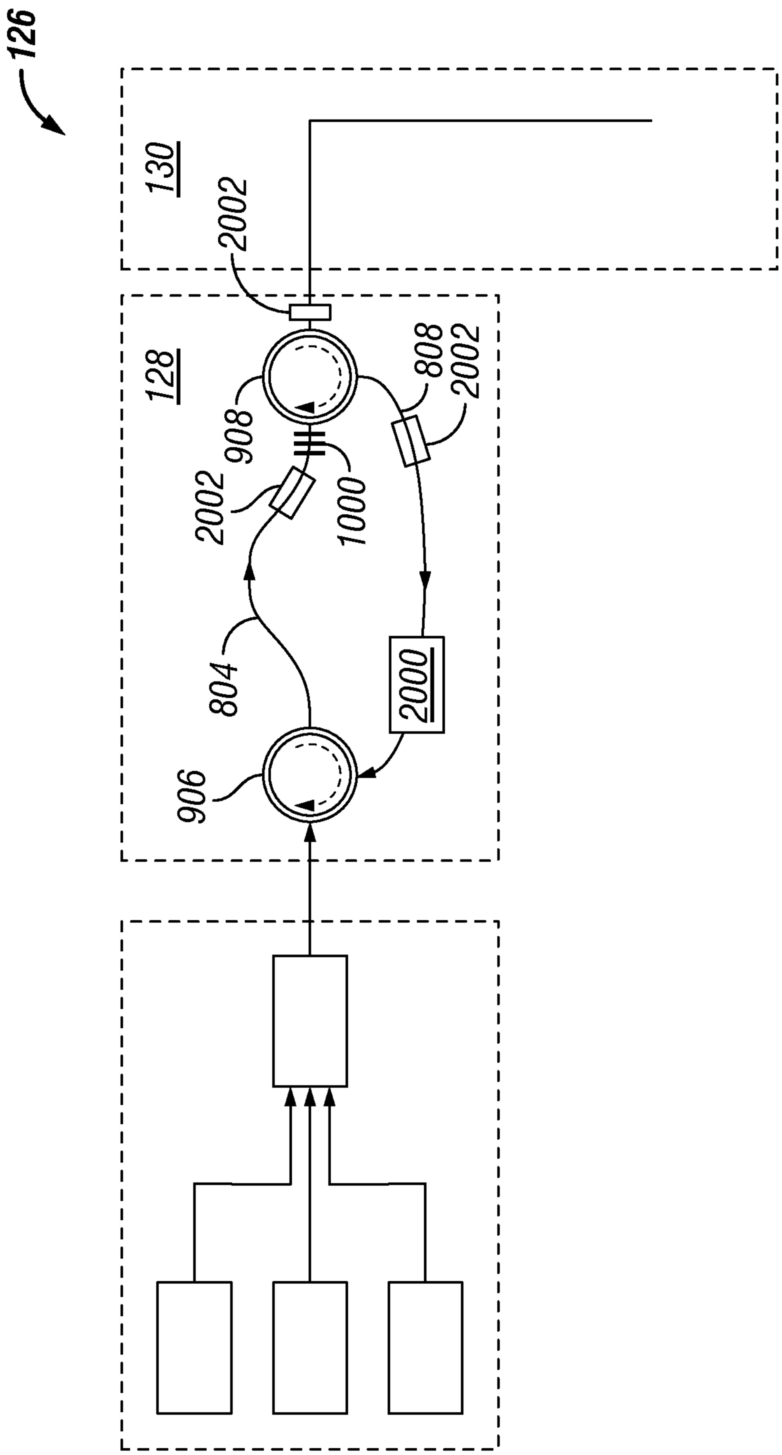
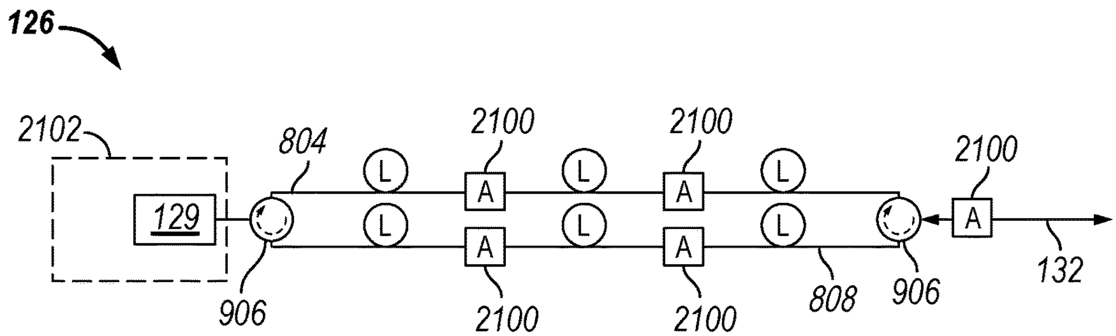


FIG. 20


126
2102
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804
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L
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2100
808
132

FIG. 21

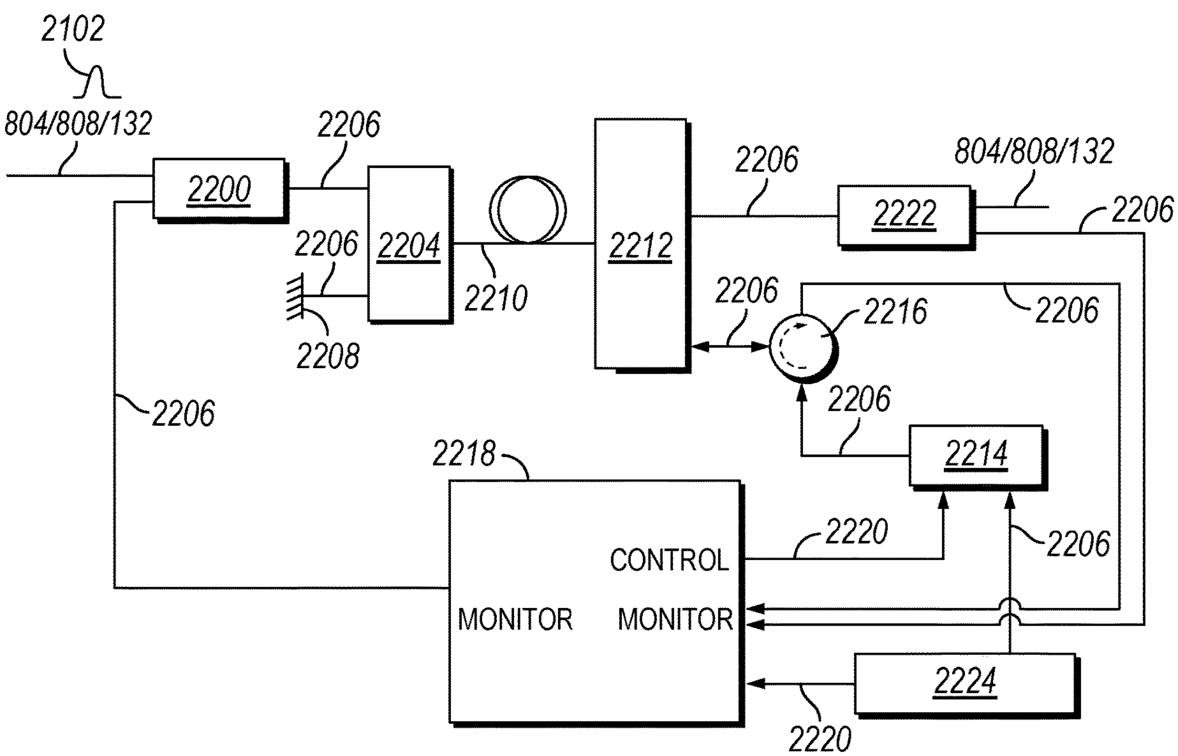
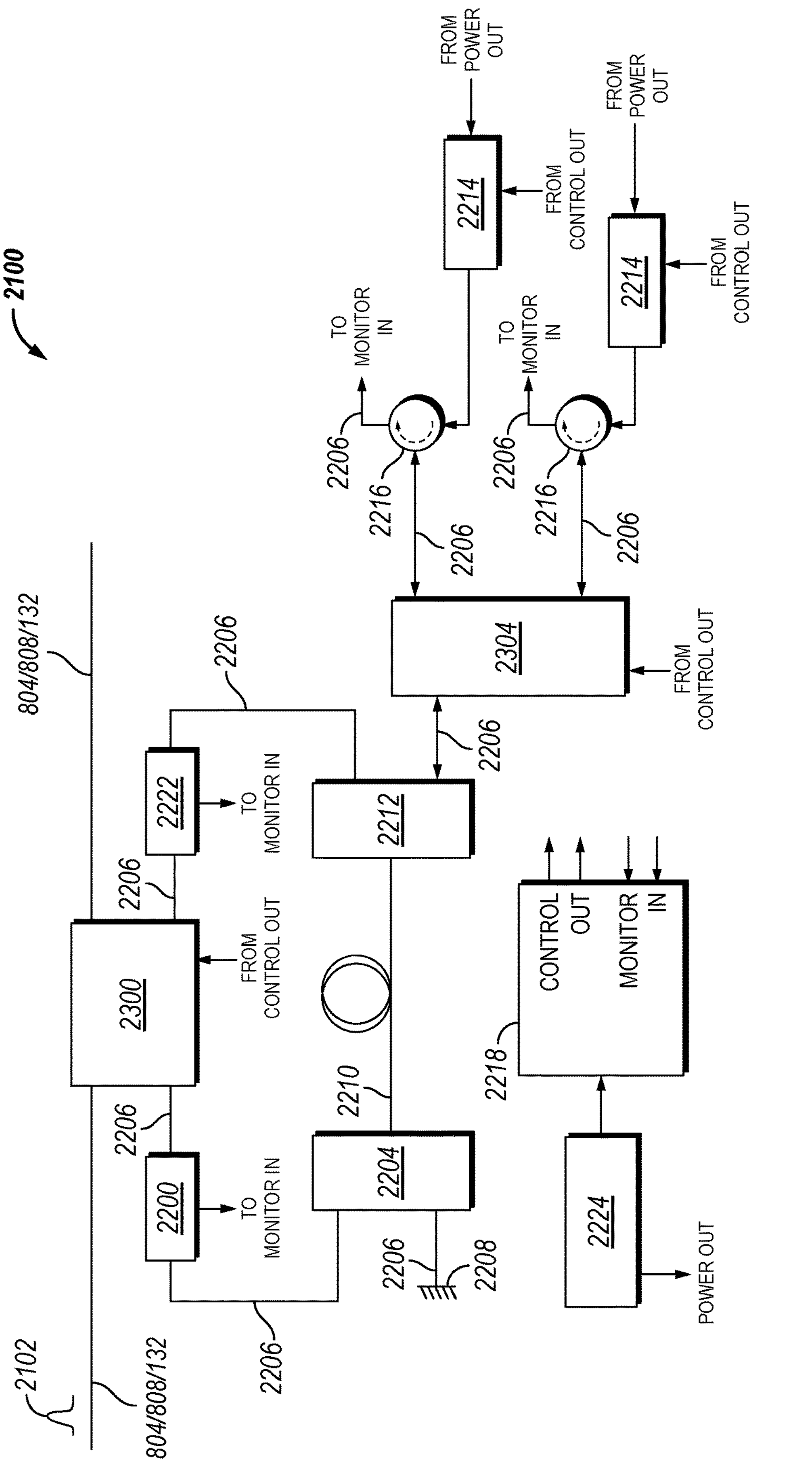

2102
804/808/132
2200
2206
2204
2208
2210
2212
2222
2216
2214
2218
CONTROL
MONITOR
MONITOR
2220
2224

FIG. 22


2100
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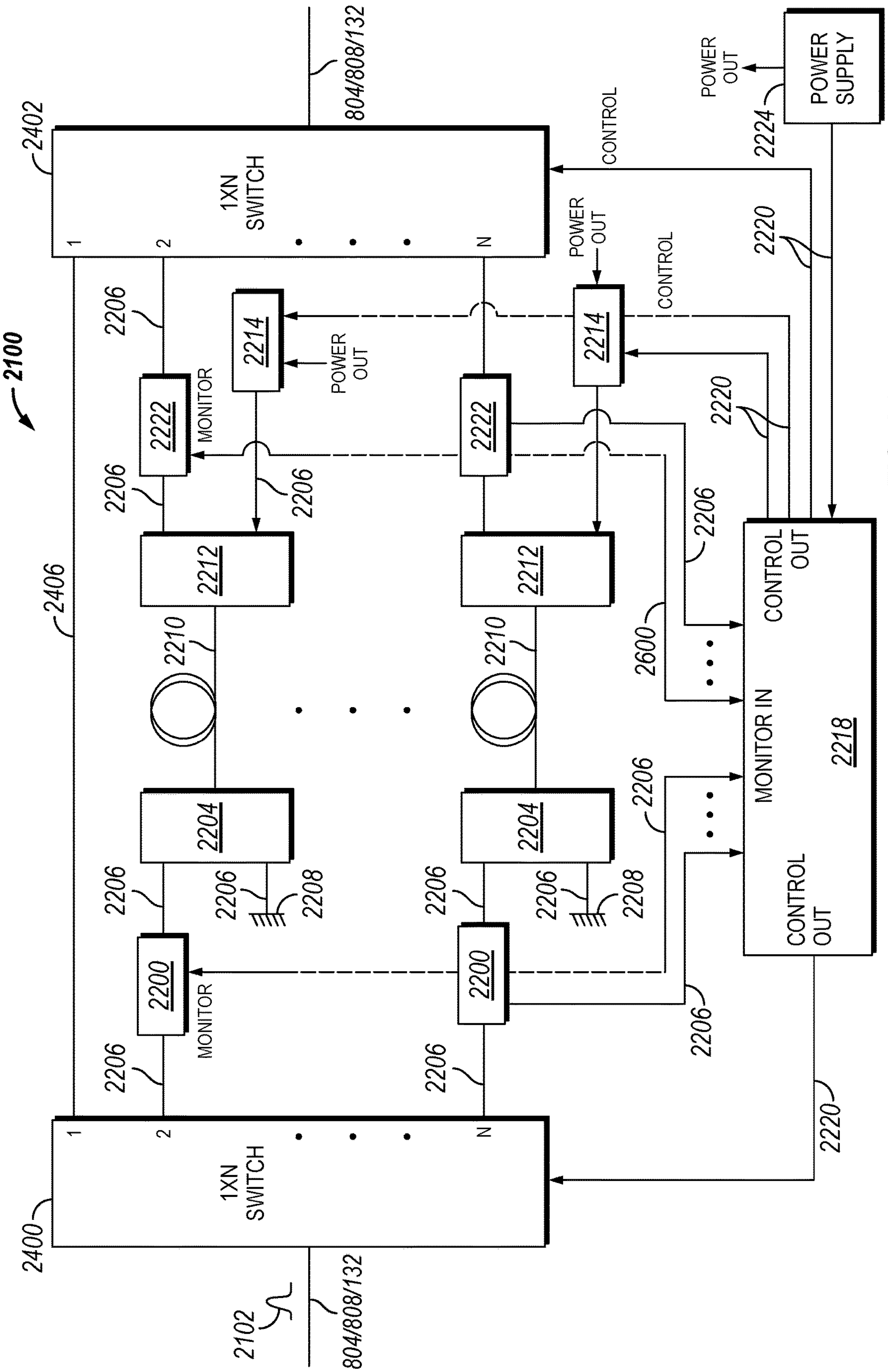
804/808/132
2200
2300
2222
2204
2212
2304
2214
2214
2216
2216
2218
2224
2206
2208
2210
TO MONITOR IN
TO MONITOR IN
TO MONITOR IN
TO MONITOR IN
FROM CONTROL OUT
FROM CONTROL OUT
FROM CONTROL OUT
FROM CONTROL OUT
FROM POWER OUT
FROM POWER OUT
CONTROL OUT
MONITOR IN
POWER OUT

FIG. 23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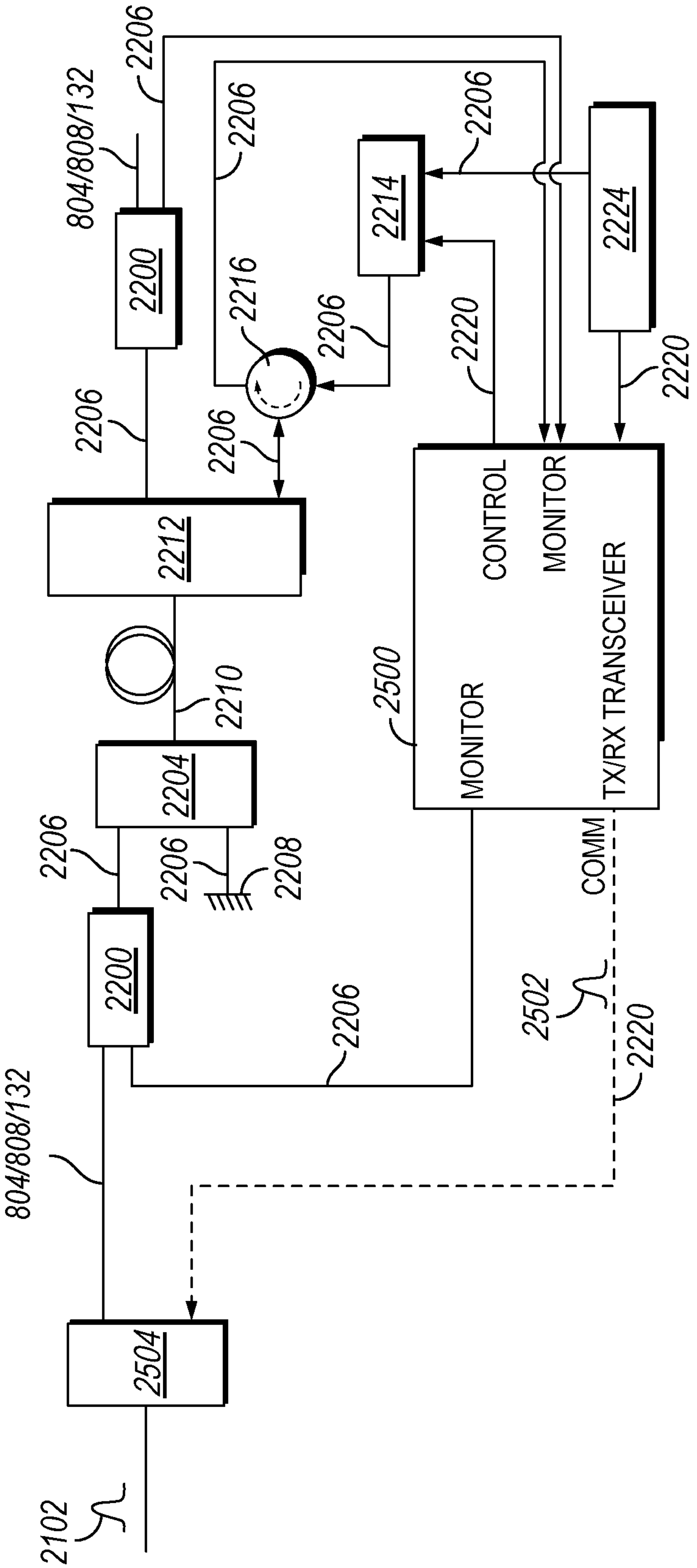
804/808/132
2402
1XN SWITCH
1
2
N
CONTROL
POWER OUT
POWER SUPPLY
2224
2220
2206
2214
POWER OUT
CONTROL
2222
MONITOR
2212
2210
2204
2208
2200
2406
2600
CONTROL OUT
MONITOR IN
2218
2400
2102

FIG. 24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2212
2206
CONTROL
MONITOR
TX/RX TRANSCEIVER
2500
MONITOR
2210
2204
2206
2206
2208
COMM
2200
2206
2502
2220
804/808/132
2504
2102

FIG. 25

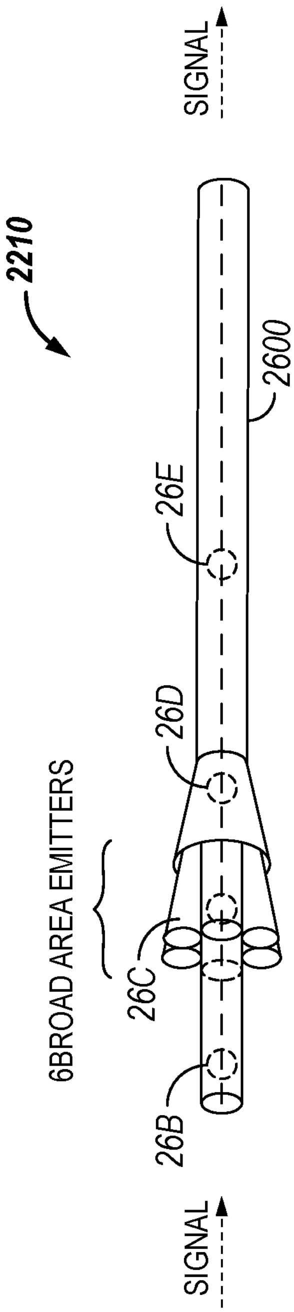

FIG. 26A

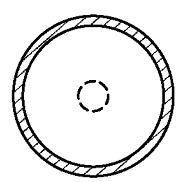

FIG. 26B

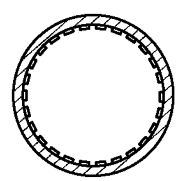

FIG. 26C

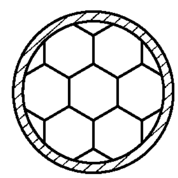

FIG. 26D

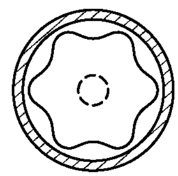

FIG. 26E

SELF-POWERED OPTICAL AMPLIFIERS

BACKGROUND

[0001] Boreholes drilled into subterranean formations may enable recovery of desirable fluids (e.g., hydrocarbons), or geological storage of other fluids (e.g., carbon dioxide), using a number of different techniques. A number of fiber optic sensing (FOS) systems and techniques may be employed in subterranean operations to characterize and monitor borehole and/or formation properties. For example, Distributed Temperature Sensing (DTS), Distributed Strain Sensing (DSS), and Distributed Acoustic Sensing (DAS) along with a fiber optic system may be utilized together to determine borehole and/or formation properties including but not limited to production profiling, solids production, injection profiling, flow assurance, vertical seismic profiling, well integrity, geological integrity, and leak detection. Distributed fiber optic sensing is a cost-effective method of obtaining real-time, high-resolution, highly accurate temperature and/or strain (static or dynamic, including acoustic) and/or pressure data along the entire wellbore. Discrete (or point) fiber optic sensing, e.g., by using fiber Bragg gratings (FBGs), is an alternative cost-effective method of obtaining real-time, high resolution, highly accurate temperature and/or strain data at discrete locations along the wellbore. Moreover, FBGs and the downhole cable may be integrated with transducers capable of inducing temperature and/or strain upon at least one FBG, thus providing an optically proportional measure of transduction, e.g., for sensing pressure, voltage, current, or chemical concentration. Additionally, fiber optic sensing may eliminate downhole electronic complexity by shifting all electrical and electro-optical systems to the surface within the interrogator unit(s). Fiber optic cables may be permanently deployed downhole in a wellbore via single- or dual-trip completion strings, behind casing, on tubing, or in pumped down installations; or temporally via coiled tubing, wireline, slickline, or disposable cables.

[0002] Distributed fiber optic sensing can be enabled by continuously sensing along the length of the optical fiber, and effectively assigning discrete measurements to a position or set of positions along the length of the fiber via optical time-domain reflectometry (OTDR). That is, by knowing the velocity of light in fiber, and by measuring the time it takes the backscattered light to return to the detector inside the interrogator, it is possible to assign a measurement and distance along the fiber. In alternative embodiments, functionally equivalent distributed fiber optic sensing data may be acquired via optical frequency-domain reflectometry (OFDR) techniques.

[0003] DAS, DSS, DTS, and FBG sensing has been practiced for monitoring downhole sensing fibers in dry Christmas tree (or dry-tree) wells to enable interventionless, time-lapse temperature, acoustic, and pressure monitoring borehole and/or formation properties including but not limited to production profiling, solids production, injection profiling, flow assurance, vertical seismic profiling, well integrity, geological integrity, and leak detection. For installation in dry-tree wells, multiple sensing fibers are typically integrated in a tubing encapsulated fiber (TEF) cable. This enables, for example, a Distributed Acoustic Sensing (DAS) system to preferentially sense a single-mode downhole sensing fiber, and a Raman-based DTS system to preferentially sense a multi-mode downhole sensing fiber; such that the DAS and DTS systems are operated simultaneously but are not simultaneously sensing the same downhole sensing fiber. Typically, the interrogator units are adjacent to, or a short distance, from the well head outlet on the dry Christmas tree.

[0004] For downhole sensing fibers installed in subsea wells, marinization of the interrogator(s) (i.e., packaging interrogators for deployment on a structure residing on the sea floor proximal to a subsea Christmas tree) introduces significant complexity and cost to the Subsea Production System (SPS) and related electrical and optical distribution systems and does not readily permit interrogator hardware upgrades. It is preferable to maintain any interrogator system (s) on the topside facility, and to sense the downhole sensing fiber through optical distribution in the subsea infrastructure. However, such a subsea well sensing operation then utilize optical engineering solutions to compensate for insertion losses accumulated through long (~5 to 100+km) lengths of subsea transmission fiber between the topside facility and subsea tree (e.g., static umbilical lines, dynamic umbilical lines, jumper cables, optical flying leads), up to 10 km of downhole sensing fiber, and multiple wet- and dry-mate optical connectors, splices, and an optical feedthrough systems (OFS) in the subsea Christmas tree (XT).

[0005] The current (horizontal or vertical) subsea XT OFS by TE Connectivity enables optical wet mating of a single fiber when the XT is landed on the tubing hanger. Thus, the number of downhole sensing fibers in a subsea well is currently limited to one. However, multi-fiber OFSs are being developed, and will enable multiple (e.g., three to six) downhole sensing fibers in a subsea well. Whether one or more downhole sensing fibers are installed in the subsea well, they utilize optical continuity back to the topsides facility so the backscattered light can be received by the interrogator(s). The optical connectors used in the subsea infrastructure (e.g., at umbilical termination assemblies, optical distribution units, drill centers, optical flying leads, and subsea trees) are finite in their optical circuit (or pin) count. For example, current wet-mate connector technology may support eight, twelve, or twenty-four pins. Also, there are physical limits (i.e., real estate) as to how many connector receptacles may be installed on subsea equipment. For example, umbilical termination assemblies (UTAs) may terminate multiple electric, hydraulic, and fiber optic cables through a finite number of electrical, hydraulic, and optical connector receptacles be mounted on their remotely operated vehicle (ROV) panels. Thus, without an efficient method to manage optical distribution, there is considerable complexity and cost when contemplating the use of multiple downhole sensing fibers in multiple subsea wells.

[0006] As the systems for subsea fiber optic sensing become more complicated and complex, extending the reach of these fiber optic sensing systems becomes costly and even more complex. Currently optical sensing technology is limited by Signal to Noise Ratio (SNR) due to optical attenuation of transmission fibers. Even extending the reach of current systems by ten kilometers, let alone hundreds of kilometers, leads to increased SNR and loss of data due to attenuation.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] These drawings illustrate certain aspects of some examples of the present disclosure and should not be used to limit or define the disclosure.

[0008] FIGS. **1**A and **1**B illustrate an example of a well measurement system in a subsea environment;

[0009] FIGS. **2**A-**2**C illustrates examples of a downhole fiber deployed in a wellbore;

[0010] FIG. **3** illustrates an optical distribution unit;

[0011] FIG. **4** illustrates an umbilical termination assembly;

[0012] FIG. **5** illustrates an optical flying lead;

[0013] FIG. **6**A illustrates an optical feedthrough system;

[0014] FIG. **6**B illustrates a cutaway of at least a part of subsea tree;

[0015] FIG. **7** illustrates an example of a FOS system;

[0016] FIG. **8** illustrate an example of a FOS system with lead lines;

[0017] FIG. **9** illustrates a schematic of another example FOS system;

[0018] FIG. **10** illustrates an example of a remote circulator arrangement;

[0019] FIG. **11** illustrates a graph for determining time for a light pulse to travel in a fiber optic cable;

[0020] FIG. **12** illustrates another graph for determining time for a light pulse to travel in a fiber optic cable;

[0021] FIG. **13** illustrates an example of a remote circulator arrangement;

[0022] FIG. **14** illustrates another graph for determining time for a light pulse to travel in a fiber optic cable;

[0023] FIG. **15**A illustrates a graph of sensing regions in the DAS system;

[0024] FIG. **15**B illustrates a graph with an active proximal circulator using an optimized FOS sampling frequency of 12.5 kHz;

[0025] FIG. **15**C illustrates a graph with a passive proximal circulator using an optimized FOS sampling frequency of 12.5 kHz;

[0026] FIG. **16** illustrates a graph of optimized sampling frequencies in the FOS system;

[0027] FIG. **17** illustrates an example of a workflow for optimizing the sampling frequencies of the DAS system; and

[0028] FIGS. **18**-**20** illustrate other examples of the FOS system;

[0029] FIGS. **21**-**25** illustrate examples of self-powered optical amplifiers; and

[0030] FIGS. **26**A-**26**E illustrate an example of an Erbium doped fiber.

DETAILED DESCRIPTION

[0031] The present disclosure relates generally to a system and method for extending the reach of fiber optic sensing, which may include but not limited to Fiber Bragg Gratings (FBGs), Self-Powered Optical Amplifiers (SPOAs), Distributed Acoustic Sensing (DAS), Distributed Temperature Sensing (DTS), Distributed Strain Sensing (DSS), Distributed Chemical Sensing (DCS), Distributed Magnetomotive Force Sensing (DMS), Distributed Electromotive Force Sensing (DES), and Distributed Brillouin-Frequency Sensing (DBFS), the latter which may be used in the extraction of distributed strain, temperature, or pressure or a combination thereof. It should be noted that any, or any combination of all systems and methods described above are generally referred to as a Fiber Optic Sensing (FOS) system. Subsea well sensing operations may present optical challenges which may relate to the signal fidelity and quality of FOS system given the long transmission fiber and multiple optical connections utilized to lead into the downhole sensing fibers. The sensing region of interest is typically the downhole sensing fiber (i.e., the in-well and reservoir sections), and not the transmission fibers (i.e., OFLs, jumpers, and static and/or dynamic umbilical lines).

[0032] To prevent a reduction in FOS signal-to-noise (SNR) and signal quality and fidelity, the FOS system described below may increase the returned signal strength with given pulse power for emitted light, decrease the noise floor of the receiving optics to detect weaker power pulses, maintain the pulse power as high as possible as it propagates along the transmission fiber(s), increase the number of light pulses that may be launched into the downhole sensing fiber(s) per second, and/or increase the maximum pulse power that may be used for given fiber length.

[0033] FOS systems utilize one or more downhole sensing fibers integrated in fiber optic cables (or tubing encapsulated fibers, TEFs). One or more electrical conductors may be integrated in the TEF so as to provide electrical power and/or telemetry to a downhole device, e.g., a pressure gauge. Downhole sensing fibers may be at least one single-mode fiber (SMF), at least one multi-mode fiber (MMF), or a combination of at least one SMF and at least one MMF. Each of the at least one SMF or MMF may be treated with a coating to prevent undesirable effects, e.g., hermetically sealed in carbon to delay hydrogen degradation. Each of at least one SMF or MMF may be treated with a coating to generate desirable effects, e.g., induced strain via improved strain transduction, a chemical reaction, or exposure to an electromotive or magnetomotive force. At least one SMF may further be enhanced (or engineered) to yield a higher-than-Rayleigh scattering coefficient so as to increase the Distributed Acoustic Sensing (DAS) signal to noise ratio (SNR) by 10 dB to 20 dB. Such enhanced backscatter fibers (EBF) may consist of either weak, distributed gratings, or discrete gratings in a SMF. The EBF may be fabricated with a narrow-enhanced backscatter bandwidth, such that a DAS system may be sensitive to the enhanced backscatter, but at least one other FOS system does not exhibit any appreciable sensitivity to the enhanced backscatter than it would if sensing a standard (or non-enhanced) SMF. The EBF may be fabricated with a broad enhanced bandwidth, such that a DAS system and at least one other FOS system may exhibit sensitivity to the enhanced backscatter.

[0034] Fiber optic cables may be permanently deployed in a subsea well via single- or dual-trip completions. Fiber optic cables may include one of at least one optical fiber encapsulated in a hydrogen-scavenging gel-filled stainless-steel tube and may further be encapsulated in a metallic (e.g., Inconel® alloy 825) armor. A hydrogen delay barrier may be located between the stainless-steel tube and the armor, e.g., a metallurgical hydrogen delay barrier such as aluminum may be extruded upon the stainless-steel tube before encapsulation in the metallic armor. The fiber optic cables may be further encapsulated in a thermoplastic encapsulation. As discussed above and below, fiber optic cables may comprise one for more fiber optic lines. Additionally, fiber optic cables may be referred to as fiber optic lines and are interchangeable for this disclosure.

[0035] FOS systems utilize transmission fibers integrated in the subsea infrastructure fiber optic cables to provide optical continuity between the interrogator(s) located at the topside facility and downhole sensing fiber(s) in the subsea well. The transmission fibers may be integrated within OFLs, jumpers, and static and/or dynamic umbilical lines, and optically coupled via splices, wet-mate connectors, and/or dry-mate connectors. Transmission fibers may be either SMF or MMF. In some embodiments, the transmission fibers may be low-loss (LL) or ultra-low loss (ULL) SMFs that have lower optical attenuation and higher power handling capability before non-linearity so as to enable high gain, co- or counter-propagating distributed Raman amplification. For example, pure silica core SMF, such as Corning® SMF-28® ULL SMF, typically exhibit 0.15 to 0.17 dB/km optical attenuation at 1550 nm wavelengths.

[0036] FOS systems may employ distributed fiber optic sensing, which is a cost-effective method of obtaining real-time, high-resolution, highly accurate temperature, strain, and acoustic/vibration data along the entire downhole fiber, while simultaneously eliminating downhole electronic complexity by shifting all electro-optical system complexity to the interrogator unit(s) located at the topside facility. The interrogator unit(s) is one or more integrated housings inclusive of the launch and receive systems of the fiber optic interrogator system. For the example of a DAS interrogator unit, the launch systems include lasers, pulsers, amplifiers, and shutter sub-systems for generating the probe pulse that enters the sensing fiber, and the receive systems include amplifiers, interferometers, digitizers, and related digital processing systems for receiving and analyzing the Rayleigh backscattered light from the sensing fiber. Example of distributed fiber optic sensing include distributed acoustic sensing (DAS), also referred to as distributed vibration sensing (DVS), which preferentially operates with SMF; distributed Brillouin-frequency sensing for distributed temperature and/or strain sensing and/or pressure sensing (DTS/DSS/DPS) preferentially operates with SMF; and Raman DTS which preferentially operates with MMF. Other distributed fiber optic sensing may include but not be limited to distributed chemical sensing (DCS), distributed electromotive force sensing (DES), and distributed magnetomotive force sensing (DMS).

[0037] Distributed fiber optic sensing may operate by continuously sensing along the length of the downhole sensing fiber, and effectively assigning discrete measurements to a position along the length of the fiber via optical time-domain reflectometry (OTDR). That is, by knowing the velocity of light in fiber, and by measuring the time it takes the backscattered light to return to the detector inside the interrogator, it is possible to assign a distance along the fiber. In alternative embodiments, functionally equivalent distributed fiber optic sensing data may be acquired via optical frequency-domain reflectometry (OFDR) techniques.

[0038] Discrete, or point, fiber optic sensing is an alternative cost-effective method of obtaining real-time, high-resolution, highly accurate temperature and/or strain (acoustic) data at discrete locations/points along the entire wellbore, while simultaneously eliminating downhole electronic complexity by shifting all electro-optical complexity to the interrogator unit(s) located at the topside facility. Point sensors may include one or more fiber Bragg gratings (FBGs), where the optical waveguide containing the FBG may be modified by a sensor assembly which efficiently transduces a measurement to temperature and/or strain upon at least one FBG. An example of such a sensor assembly is a pressure and temperature gauge, a chemical sensor, and a voltage sensor. FBGs may operate with either SMF or MMF.

[0039] The subsea well's downhole sensing fiber connects to the subsea optical distribution system via an optical feedthrough system (OFS) in the subsea Christmas tree (XT) and tubing hanger. The XT may be either a vertical (VXT) or a horizontal XT (HXT) design, or any hybrid or simplified solution where to hang off the downhole completions. The methods and systems described below are agnostic to the use of VXTs or HXTs. In the following description, VXT, HXT, subsea Christmas tree, wet Christmas tree, wet-tree, and subsea tree are all synonymous. The OFS provides optical continuity from transmission fibers in the subsea optical distribution system to the downhole sensing fiber via an assembly of wet- and dry-mate optical connectors and/or splices. When the XT is landed on the tubing hanger, the OFS enables at least one fiber to be optically continuous between the XT's ROV panel and the tubing hanger. Current and future OFS products from TE Connectivity and Teledyne enable at most one, three, or six fibers to be fed through the XT. Fibers may be SMF, MMF, or any combination of SMF and MMF.

[0040] From a downhole monitoring system consideration, multiple downhole fibers may increase data acquisition opportunities while simplifying overall downhole monitoring system complexity. For example, one SMF may be used for acquiring DAS and/or DTS, and two SMFs may each or both be used for FBG sensing arrays of pressure and temperature gauges. For intelligent completions, this may potentially eliminate the necessity of electric pressure and temperature gauge arrays, and thus simplify subsea control and power distribution systems. The challenge is that having multiple downhole sensing fibers with their necessity for optical continuity back to the interrogators located at the topside facility, which could place significant complexity, burden, and cost on the subsea optical distribution system. On a per-well basis, the systems and methods described below may maximize the number of downhole sensing fibers while minimizing the number of subsea transmission fibers needed for their continuity from XT to the topside facility.

[0041] The subsea optical distribution system provides optical continuity from the downhole sensing fiber to the interrogator located at the topside facility. The optical distribution system may be stand-alone (separated) or integrated with other (e.g., electric and/or hydraulic) utilities of the subsea production system (SPS). This may involve multiple optical flying leads (OFLs), jumper cables, static umbilical lines, dynamic umbilical lines, subsea umbilical termination assemblies (SUTAs), topside umbilical termination assemblies (TUTAs), surface cables between the TUTAs and interrogator(s), optical distribution units (ODUs), and optical distribution through drill centers, manifold centers, or other subsea equipment.

[0042] Systems and methods discussed below may also be utilized to extend the reach of fiber optic sensing systems. Specifically, the use of localized power generation nodes (SPOAs) that may allow for the extension of sensing without increasing the burden to existing SUFT infrastructure with additional electrical and/or optical circuits. This may allow for signal simplification as necessary to meet standards for transmitting signals over extreme distances without increasing SNR or the loss of signal from attenuation. This may be performed by stand-alone generation nodes that may enable signal amplification within the extending reach sensing system thus enabling extreme extended reach sensing.

[0043] FIGS. **1**A and **1**B illustrate an example of a well system **100** that may employ the principles of the present disclosure. More particularly, well system **100** may include a floating vessel **102** centered over a subterranean hydrocarbon bearing formation **104** located below a sea floor **106**. As illustrated, floating vessel **102** is depicted as an offshore, semi-submersible oil and gas drilling platform, but could alternatively include any other type of floating vessel such as, but not limited to, a drill ship, a pipe-laying ship, a tension-leg platforms (TLPs), a spar platform, a production platform, a floating production, storage, and offloading (FPSO) vessel, a floating production unit (FPU), and/or the like. Additionally, and without loss of generality, the methods and systems described below may also be utilized for subsea tie-backs to a fixed offshore platform, an onshore facility, or a facility on an artificial island. Moreover, the systems and methods of the present disclosure are applicable to onshore reservoirs and related facilities. A subsea conduit or riser **108** extends from a deck **110** of floating vessel **102** to sea floor **106** and may connect to a production manifold **112**. As illustrated, static pipe **114** may run from production manifold **112** to a pipeline end termination **116**. Flexible pipe **118** may attach a subsea tree **120** to pipeline end termination **116**. In examples, flexible pipe **118** may travers from production manifold **112** and connect directly to subsea tree **120**. Additionally, flexible pipe **118** may connect one subsea tree **120** to another subsea tree **120**, effectively tying one or more subsea trees **120** together and allow for a single flexible pipe **118** to connect one or more subsea trees **120** to a single production manifold **112**.

[0044] Subsea tree **120** may cap a wellbore **122** that has been drilled into formation **104**. Within wellbore may be a completion system consisting of one or more tubulars **124** that are connected to subsea tree **120**. During operations, formation fluids may be produced from formation **104**, and flow through one or more tubulars **124** to subsea tree **120**. As subsea tree **120** is attached to floating vessel **102**, formation fluid may flow from subsea tree **120**, through flexible pipe **118**, pipeline end termination **116**, static pipe **114**, production manifold **112**, and up through riser **108** to floating vessel **102** for processing, storage, and subsequent offloading or export.

[0045] To monitor downhole operations, a Fiber Optic Sensing (FOS) system **126** may be employed from floating vessel **102**. FOS **126** system utilizes distributed and/or discrete fiber optic sensing as a cost-effective method of obtaining real-time, high-resolution, highly accurate physical measurements, such as but not limited to temperature, strain, and acoustic measurements along the entire wellbore, while simultaneously eliminating downhole electronic complexity by shifting all electro-optical complexity to the interrogator unit (IU), also called an interrogator, located onboard the floating vessel **102**. FOS system **126** may include an interrogator **128**, umbilical line **130**, and at least one downhole sensing fiber **132**. As illustrated, interrogator **128** may be at least partially disposed on floating vessel **102**. Interrogator unit **128** may connect to umbilical line **130**. Umbilical line **130** may include one or more optical fibers that traverse from a local electronics room (LER) or central control room (CCR) to a topside umbilical termination assembly (TUTA) onboard floating vessel **102**. Umbilical line **130** may include a dynamic umbilical line **134**, a subsea umbilical termination assembly (SUTA) **140**, and a static umbilical line **136**. Umbilical line may further include optical flying lead **142** and optical feedthrough system **144**. Umbilical line **130** may include one or more fiber optic cables. Each fiber optic cable may include one or more optical fibers.

[0046] FIG. **3** illustrates an optical distribution unit **138**. As illustrated, one of ordinary skill in the art may recognize that optical distribution unit **138** may be constructed to withstand pressures, temperatures, and a subsea environment in which optical distribution unit **138** may operate and function. During operations, a remotely operated vehicle (ROV) (not illustrated) may be deployed from vessel **102** or another vessel with optical distribution unit **138**. The ROV may place optical distribution unit **138** in a previously designated area on sea floor **106**. Once deployed, optical distribution unit **138** may act as a terminal in which dynamic umbilical line **134** of umbilical line **130** attaches to from vessel **102** (e.g., referring to FIGS. **1**A and **1**B). One or more ROVs may be utilized to attach dynamic umbilical line **134** and static umbilical line **136** to optical distribution unit **138**. Additionally, this procedure, in some operations, may be performed at surface **103** on vessel **102**. In examples, one or more dynamic umbilical lines **134** may attach to one or more input connectors **300**. This may allow for one or more static umbilical lines **136** to connect to one or more output connectors **302**. Thus, one or more static umbilical lines **136** may allow for a single vessel **102** to service one or more subsea trees **120** that are connected to optical distribution unit **138**. To reach subsea trees **120**, one or more static umbilical lines **136** traverse to one or more umbilical termination assemblies **140**. Additionally, in examples, a flying optical lead **142** (discussed below) may be utilized to connect optical distribution unit **138** to one or more subsea trees **120**.

[0047] FIG. **4** illustrates an umbilical termination assembly **140**. As illustrated, one of ordinary skill in the art may recognize that umbilical termination assembly **140** may be constructed to withstand pressures, temperatures, and a subsea environment in which umbilical termination assembly **140** may operate and function. During operations, one or more ROVs (not illustrated) may be deployed from vessel **102** or another vessel with umbilical termination assembly **140**. The ROV may place umbilical termination assembly **140** in a previously designated area on sea floor **106**. Once deployed, umbilical termination assembly **140** may act as a terminal in which static umbilical line **136** of umbilical line **130** attaches to from optical distribution unit **138** (e.g., referring to FIGS. **1**A and **1**B). One or more ROVs may be utilized to attach static umbilical line **136** to umbilical termination assembly **140**. Additionally, this procedure, in some operations, may be performed at surface **103** on vessel **102**. In examples, one or more dynamic umbilical lines **134** may attach to one or more input connectors **300**. From umbilical termination assembly **140**, an optical flying lead **142** may connect umbilical termination assembly **140** at one or more output connectors **302** to an optical feedthrough system **144** that is disposed in or is at least a part of subsea tree **120** (e.g., referring to FIGS. **1**A and **1**B).

[0048] FIG. **5** illustrates an optical flying lead. An optical flying lead **142** is a flexible connection that may attach optical distribution unit **138** or umbilical termination assembly **140** or any other suitable location in the optical distribution system to optical feedthrough system **144**. As illustrated, optical flying lead **142** includes a flexible hose **500** terminated at both ends with optical wet-mate connectors **504**. Flexible hose **500** includes one or more optical fibers that provide optical continuity between the two optical wet-mate connectors **504**. Flexible hose **500** may be filled with fluid for pressure balancing in subsea environments. Additionally, an integrated compartment **502** may be disposed at any distance along the flexible hose **500**. Integrated compartment **502** may include any number of optical devices, which is discussed in detail below. Integrated compartment **502** may be rated as a one atmosphere (1 atm) pressure cannister qualified for deployment in subsea environments and may contain a nitrogen-purged atmospheric environment. Each optical wet-mate connection **504** is configured to allow for an ROV to attach optical flying lead **142** to optical feedthrough system **144** and optical distribution unit **138** or umbilical termination assembly **140** or any other suitable location in the optical distribution system, as is readily understood to those of ordinary skill in the art.

[0049] FIG. **6**A illustrates a subsea tree **120** with optical feedthrough system **144**. As illustrated, one of ordinary skill in the art may recognize that subsea tree **120** with optical feedthrough system **144** may be constructed to withstand pressures, temperatures, and a subsea environment in which subsea tree **120** and optical feedthrough system **144** may operate and function. During manufacturing of subsea tree **120**, optical feedthrough system **144** may be integrated into subsea tree **120** and tubing hanger assemblies. Subsea tree **120** and tubing hanger assemblies each contain an optical wet-mate receptacle **600** (e.g., referring to FIG. **6**B) that may be optically coupled when subsea tree **120** and tubing hangers are operationally deployed. During installation operations, the tubing hanger assembly is coupled to the upper completion of wellbore **122** with optical continuity to downhole sensing fiber **132** (e.g., referring to FIGS. **1**A and **1**B), and landed into wellbore **122** on sea floor **106** (e.g., referring to FIGS. **1**A and **2**B). Subsea tree **120** is then landed upon the tubing hanger such that subsea tree **120** and tubing hanger are optically coupled via the mated optical wet-mate receptacle **600**. One or more ROVs may be utilized to attach optical flying lead **142** (e.g., referring to FIGS. **1**A and **1**B) to optical wet-mate receptacle **602** located on the ROV panel **604** of subsea tree **120** and optical feedthrough system **144** as well as optical distribution unit **138** or umbilical termination assembly **140**. In examples, one or more static umbilical lines **136** may attach directly to subsea trees **120** without optical flying lead **142**. Subsea tree **120** and optical feedthrough system **144** may allow for optical flying lead **142** and/or one or more static umbilical lines **136** to connect to one or more downhole sensing fibers **132**.

[0050] FIG. **6**B illustrates optical feedthrough system **144** formed when subsea tree **120** (e.g., referring to FIG. **6**A) has been landed upon a tubing hanger. In examples, optical flying lead **142** may attach optical wet-mate receptacle **602** located on ROV panel **604** of subsea tree **120** (e.g., referring to FIGS. **6**A), which is connected to a pressure-compensated flexible hose **606** that terminates with a an optical dry-mate connection **610** at subsea tree block **608**. Optical dry-mate connection **600** is connected to the subsea tree's optical wet-mate receptacle **600**. During installation operations, subsea tree **120** is landed upon the tubing hanger such that subsea tree's optical wet-mate receptacle **600** optically connects to tubing hanger's optical wet-mate receptacle **612**. In some embodiments, the tubing hanger's optical wet-mate receptacle **612** is connected to an optical dry-mate receptacle **614** at the base of the tubing hanger, and optically connected to a pigtail **618** with optical dry-mate receptacle **616**. Pigtail **618** is connected to downhole sensing fiber **132** via a splice assembly **620** in the upper completion. In other embodiments, tubing hanger's optical wet-mate receptacle **612** is optically connected to downhole sensing fiber **132** via a splice assembly **620** in the upper completion. Although not illustrated, one or more downhole sensing fibers **132** may be disposed in a fiber optic cable that is optically connected to tubing hanger's optical wet-mate receptacle **612**. In examples, an integrated compartment **502** may be installed along flexible hose **606** between subsea tree's ROV panel **604** and the optical dry-mate connection **600** at subsea tree block **608**. This integrated compartment may include any number of optical devices, which is discussed in detail below. Integrated compartment **502** may be a one atmosphere (1 atm) pressure cannister rated for deployment in subsea environments and may contain a nitrogen-purged atmospheric environment. As illustrated, and discussed below in further detail, optical feedthrough system **144** allows for optical coupling between optical flying lead **142** and one or more downhole sensing fibers **132** through a single connection. As will be discussed in more detail below, downhole sensing fibers **132** may allow for downhole measurements to be taken within wellbore **122** utilizing principles and function associated with FOS **126**.

[0051] Referring back to FIGS. **1**A and **1**B, wellbore **122** extends through the various earth strata toward the subterranean hydrocarbon bearing formation **104** and tubular **124** may be extended within wellbore **122**. Even though FIGS. **1**A and **1**B depict a vertical wellbore **122**, it should be understood by those skilled in the art that the methods and systems described are equally well suited for use in horizontal or deviated wellbores. During drilling operations, a drill sting, may include a bottom hole assembly (BHA) that includes a drill bit and a downhole drilling motor, also referred to as a positive displacement motor ("PDM") or "mud motor." During production operations, the completion system represented by tubular **124** may include one or more downhole sensing fibers **132** of a FOS system **126**.

[0052] Downhole sensing fiber **132** may be permanently deployed in a wellbore via single- or dual-trip completion systems, behind casing, on tubing, or in pumped down installations. FIGS. **2**A-**2**C illustrate examples of different types of deployment of downhole sensing fiber **132** in wellbore **122** (e.g., referring to FIGS. **1**A and **1**B). As illustrated in FIG. **2**A, wellbore **122** deployed in formation **104** may include surface casing **200** in which production casing **202** may be deployed. Additionally, production tubing **204** may be deployed within production casing **202**. In this example, downhole sensing fiber **132** may be permanently deployed in a completion system. In examples, downhole sensing fiber **132** is attached to the outside of production tubing **204** by one or more cross-coupling protectors **210**. Without limitation, cross-coupling protectors **210** may be evenly spaced and may be disposed on every other joint of production tubing **204**. Further illustrated, downhole sensing fiber **132** may be coupled to a fiber connection **206**. Without limitation, fiber connection **206** may attach downhole sensing fiber **132** to optical feedthrough system **144**, and/or umbilical line **130** (e.g., referring to FIGS. **1**A and **1**B) in the manner, systems, and/or methods described above. In examples, downhole sensing fiber **132** may further be optically connected to umbilical line **130** through optical flying lead **142** (e.g., referring to FIGS. **1**A and **1**B). Fiber connection **206** may operate as an optical feedthrough system **144** (itself comprising a series of wet- and dry-mate optical connectors and splices) in the wellhead that optically connects downhole sensing fiber **132** from the tubing hanger to umbilical line **130** on the subsea tree's ROV panel **604** (e.g., referring to FIGS. **6**A and **6**B). Umbilical line **130** may include to an optical flying lead **142** and may further include an optical distribution system(s) **138**, umbilical termination unit(s) **140**, and transmission fibers encapsulated in flying optical leads **142**, flow lines, rigid risers, flexible risers, and/or one or more static and/or dynamic umbilical lines. This may allow for umbilical line **130** to connect and disconnect from downhole sensing fiber **132** while preserving optical continuity between the umbilical line **130** and the downhole sensing fiber **132**.

[0053] FIG. **2**B illustrates an example of permanent deployment of downhole sensing fiber **132**. As illustrated in wellbore **122** deployed in formation **104** may include surface casing **200** in which production casing **202** may be deployed. Additionally, production tubing **204** may be deployed within production casing **202**. In examples, downhole sensing fiber **132** is attached to the outside of production casing **202** by one or more cross-coupling protectors **210**. Without limitation, cross-coupling protectors **210** may be evenly spaced and may be disposed on every other joint of production tubing **204**. downhole sensing fiber **132**

[0054] FIG. **2**C illustrates an example of a pump-down fiber operation in which downhole sensing fiber **132** may be deployed cither permanently or temporarily. As illustrated in FIG. **2**C, wellbore **122** deployed in formation **104** may include surface casing **200** in which production casing **202** may be deployed. Additionally, capillary tubing **212** may be deployed within production casing **202**. In this example, downhole sensing fiber **132** may be permanently or temporarily deployed via a pumping operation into the capillary tube.

[0055] Referring back to FIGS. **1**A and **1**B, interrogator unit **128** may be connected to an information handling system **146** through connection **148**, which may be wired and/or wireless. It should be noted that both information handling system **146** and interrogator unit **128** are disposed on floating vessel **102**. Both systems and methods of the present disclosure may be implemented, at least in part, with information handling system **146**. Information handling system **146** may include any instrumentality or aggregate of instrumentalities operable to compute, estimate, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system **146** may be a processing unit **150**, a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. Information handling system **146** may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM, and/or other types of nonvolatile memory. Additional components of the information handling system **146** may include one or more disk drives, one or more network ports for communication with external devices as well as an input device **152** (e.g., keyboard, mouse, etc.) and video display **154**. Information handling system **146** may also include one or more buses operable to transmit communications between the various hardware components.

[0056] Alternatively, systems and methods of the present disclosure may be implemented, at least in part, with non-transitory computer-readable media **156**. Non-transitory computer-readable media **156** may include any instrumentality or aggregation of instrumentalities that may retain data and/or instructions for a period of time. Non-transitory computer-readable media **156** may include, for example, storage media such as a direct access storage device (e.g., a hard disk drive or floppy disk drive), a sequential access storage device (e.g., a tape disk drive), compact disk, CD-ROM, DVD, RAM, ROM, electrically erasable programmable read-only memory (EEPROM), and/or flash memory; as well as communications media such as wires, optical fibers, microwaves, radio waves, and other electromagnetic and/or optical carriers; and/or any combination of the foregoing.

[0057] Production operations in a subsea environment may present optical challenges for a DAS based FOS system **126**. For example, in a DAS system, a maximum pulse power that may be used is approximately inversely proportional to fiber length due to optical non-linearities in the fiber. Therefore, the quality of the overall signal is poorer with a longer fiber than a shorter fiber. This may impact any FOS system **126** that may utilize DAS, since the distal end of the downhole sensing fiber **132** may include an interval of interest (i.e., the reservoir) in which the downhole sensing fiber **132** may be deployed. The interval of interest may include wellbore **122** and formation **104**. For pulsed DAS systems, in FOS system **126**, such as the one exemplified in FIG. **7**, an additional challenge is the drop-in signal to noise ratio (SNR) and spectral bandwidth associated with the decrease in the number of light pulses that may be launched into the fiber per second (i.e., DAS pulse repetition rate) when interrogating fibers with overall lengths exceeding 10 km. As such, utilizing DAS system in FOS system **126** in a subsea environment may have to increase the returned signal strength with given pulse power, increase the maximum pulse power that may be used for given fiber optic cable length, maintain the pulse power as high as possible as it propagates down the fiber optic cable length, and increase the number of light pulses that may be launched into the fiber optic cable per second.

[0058] FIG. **7** illustrates an example of DAS system for FOS **126**. The DAS system may include information handling system **146** that is communicatively coupled to interrogator unit **128**. Without limitation, DAS system may include a coherent Rayleigh scattering system with a compensating interferometer. In examples, the DAS system may be used for phase-sensitive sensing of events in a wellbore using measurements of coherent Rayleigh backscatter and/or may interrogate a downhole sensing fiber containing an array of partial reflectors, for example, fiber Bragg gratings.

[0059] As illustrated in FIG. **7**, interrogator unit **128** may include a pulse generator **700** coupled to a first coupler **702** using an optical fiber **704**. Pulse generator **700** may be a laser, or a laser connected to at least one amplitude modulator, or a laser connected to at least one switching amplifier, i.e., semiconductor optical amplifier (SOA). First coupler **702** may be a traditional fused type fiber optic splitter, a circulator, a PLC fiber optic splitter, or any other type of splitter known to those with ordinary skill in the art. Pulse generator **700** may be coupled to optical gain elements (not shown) to amplify pulses generated therefrom. Example optical gain elements include, but are not limited to, Erbium Doped Fiber Amplifiers (EDFAs) or Semiconductor Optical Amplifiers (SOAs).

[0060] FOS system **126**, which is a DAS system, may include an interferometer **706**. Without limitations, interferometer **706** may include a Mach-Zehnder interferometer. For example, a Michelson interferometer or any other type of interferometer **706** may also be used without departing from the scope of the present disclosure. Interferometer **706** may include a top interferometer arm **708**, a bottom interferometer arm **710**, and a gauge **712** positioned on bottom interferometer arm **710**. Interferometer **706** may be coupled to first coupler **702** through a second coupler **714** and an optical fiber **716**. Interferometer **706** further may be coupled to a photodetector assembly **718** of the DAS system through a third coupler **720** opposite second coupler **714**. Second coupler **714** and third coupler **720** may be a traditional fused type fiber optic splitter, a PLC fiber optic splitter, or any other type of optical splitter known to those with ordinary skill in the art. Photodetector assembly **718** may include associated optics and signal processing electronics (not shown). Photodetector assembly **718** may be a semiconductor electronic device that uses the photoelectric effect to convert light to electricity. Photodetector assembly **718** may be an avalanche photodiode or a pin photodiode but is not intended to be limited to such.

[0061] When operating FOS system **126**, pulse generator **700** may generate a first optical pulse **722** which is transmitted through optical fiber **704** to first coupler **702**. First coupler **702** may direct first optical pulse **722** through a sensing fiber **724**. It should be noted that sensing fiber **724** may be disposed in umbilical line **130** and is at least a part of downhole sensing fiber **132** (e.g., referring to FIGS. **1**A and **1**B). As illustrated, sensing fiber **724** may be coupled to first coupler **702**. As first optical pulse **722** travels through sensing fiber **724**, imperfections in sensing fiber **724** may cause a portion of the light to be backscattered along fiber optical cable **724** due to Rayleigh scattering. In other embodiments, the sensing fiber **724** may be enhanced (or engineered) to yield a higher-than-Rayleigh backscatter coefficient. Scattered light according to Rayleigh scattering is returned from every point along sensing fiber **724** along the length of sensing fiber **724** and is shown as backscattered light **726** in FIG. **7**. This backscatter effect may be referred to as Rayleigh backscatter. Density fluctuations in sensing fiber **724** may give rise to energy loss due to the scattered light, α_{scat}, with the following coefficient:

$$\alpha_{scat} = \frac{8\pi^3}{3\lambda^4} n^8 p^2 k T_f \beta \tag{1}$$

where n is the refraction index, p is the photoelastic coefficient of sensing fiber **724**, k is the Boltzmann constant, and β is the isothermal compressibility. T_f is a fictive temperature, representing the temperature at which the density fluctuations are "frozen" in the material. Fiber optical cable **724** may be terminated with a low reflection device (not shown). In examples, the low reflection device (not shown) may be a fiber coiled and tightly bent to violate Snell's law of total internal reflection such that all the remaining energy is sent out of sensing fiber **724**.

[0062] Backscattered light **726** may travel back through sensing fiber **724**, until it reaches second coupler **714**. First coupler **702** may be coupled to second coupler **714** on one side by optical fiber **716** such that backscattered light **726** may pass from first coupler **702** to second coupler **714** through optical fiber **716**. Second coupler **714** may split backscattered light **726** based on the number of interferometer arms so that one portion of any backscattered light **726** passing through interferometer **706** travels through top interferometer arm **708** and another portion travels through bottom interferometer arm **710**. Therefore, second coupler **714** may split the backscattered light from optical fiber **716** into a first backscattered pulse and a second backscattered pulse. The first backscattered pulse may be sent into top interferometer arm **708**. The second backscattered pulse may be sent into bottom interferometer arm **710**. These two portions may be re-combined at third coupler **720**, after they have exited interferometer **706**, to form an interferometric signal.

[0063] Interferometer **706** may facilitate the generation of the interferometric signal through the relative phase shift variations between the light pulses in top interferometer arm **708** and bottom interferometer arm **710**. Specifically, gauge **712** may cause the length of bottom interferometer arm **710** to be longer than the length of top interferometer arm **708**. With different lengths between the two arms of interferometer **706**, the interferometric signal may include backscattered light from two positions along sensing fiber **724** such that a phase shift of backscattered light between the two different points along sensing fiber **724** may be identified in the interferometric signal. The distance between those points L may be half the length of the gauge **712** in the case of a Mach-Zehnder configuration, or equal to the gauge length in a Michelson interferometer configuration.

[0064] While FOS system **126** is running, the interferometric signal will typically vary over time. The variations in the interferometric signal may identify strains in sensing fiber **724** that may be caused, for example, by seismic energy. By using the time of flight for first optical pulse **722**, the location of the strain along sensing fiber **724** and the time at which it occurred may be determined. If sensing fiber **724** is positioned within a wellbore, the locations of the strains in sensing **724** may be correlated with depths in the formation in order to associate the seismic energy with locations in the formation and wellbore.

[0065] To facilitate the identification of strains in sensing fiber **724**, the interferometric signal may reach photodetector assembly **718**, where it may be converted to an electrical signal. The photodetector assembly may provide an electric signal proportional to the square of the sum of the two electric fields from the two arms of the interferometer. This signal is proportional to:

$$P(t) = P1 + P2 + 2 * \sqrt{(P1P2)}\cos(\phi1 - \phi2) \tag{2}$$

where P_n is the power incident to the photodetector from a particular arm (1 or 2) and ϕ_n is the phase of the light from the particular arm of the interferometer. Photodetector assembly **718** may transmit the electrical signal to information handling system **146**, which may process the electrical signal to identify strains within sensing fiber **724** and/or convey the data to a display and/or store it in computer-readable media. Photodetector assembly **718** and information handling system **146** may be communicatively and/or mechanically coupled. Information handling system **146** may also be communicatively or mechanically coupled to pulse generator **700**.

[0066] Modifications, additions, or omissions may be made to FIG. **7** without departing from the scope of the present disclosure. For example, FIG. **7** shows a particular configuration of components of a DAS system, which is a FOS system **126**, operating via optical time-domain reflectometry (OTDR). However, any suitable configurations of components may be used, including that the DAS system may be operated via optical frequency-domain interferometry (OFDR). As another example, pulse generator **700** may generate a multitude of coherent light pulses, optical pulse **722**, operating at distinct frequencies that are launched into the sensing fiber **724** cither simultaneously or in a staggered fashion. For example, the photo detector assembly is expanded to feature a dedicated photodetector assembly for each light pulse frequency. In examples, a compensating interferometer may be placed in the launch path (i.e., prior to traveling down sensing fiber **724**) of the interrogating pulse to generate a pair of pulses that travel down sensing fiber **724**. In examples, interferometer **706** may not be necessary to interfere the backscattered light from pulses prior to being sent to photo detector assembly. In one branch of the compensation interferometer in the launch path of the interrogating pulse, an extra length of fiber not present in the other branch (a gauge length similar to gauge **712** of FIG. **7**) may be used to delay one of the pulses. To accommodate phase detection of backscattered light using FOS system **126**, one of the two branches may include an optical frequency shifter (for example, an acousto-optic modulator) to shift the optical frequency of one of the pulses, while the other may include a gauge. This may allow using a single photodetector receiving the backscatter light to determine the relative phase of the backscatter light between two locations by examining the heterodyne beat signal received from the mixing of the light from different optical frequencies of the two interrogation pulses.

[0067] In examples, the DAS system, which is a FOS system **126**, may generate interferometric signals for analysis by the information handling system **146** without the use of a physical interferometer. For instance, the DAS system may direct backscattered light to photodetector assembly **718** without first passing it through any interferometer, such as interferometer **706** of FIG. **7**. Alternatively, the backscattered light from the interrogation pulse may be mixed with the light from the laser originally providing the interrogation pulse. Thus, the light from the laser, the interrogation pulse, and the backscattered signal may all be collected by photodetector assembly **718** and then analyzed by information handling system **146**. The light from each of these sources may be at the same optical frequency in a homodyne phase demodulation system or may be different optical frequencies in a heterodyne phase demodulator. This method of mixing the backscattered light with a local oscillator allows measuring the phase of the backscattered light along the fiber relative to a reference light source.

[0068] FIG. **8** illustrates an example of DAS system, which is FOS system **126**, which may be utilized to overcome challenges presented by a subsea environment. FOS system **126** may comprise interrogator unit **128**, umbilical line **130**, and downhole sensing fiber **132**. As illustrated, interrogator unit **128** may comprise pulse generator **700** and photodetector assembly **718**, both of which may be communicatively coupled to information handling system **146**. Additionally, interferometers **706** may be placed within interrogator unit **128** and operate and/or function as described above. FIG. **8** illustrates an example of FOS system **126** in which lead lines **800** may be used. As illustrated, an optical fiber **704** may attach pulse generator **700** to an output **802**, which may be a fiber optic connector. Umbilical line **130** may attach to output **802** with a first fiber optic cable **804**. First fiber optic cable **804**, which may be referred to as a down going fiber, may traverse the length of umbilical line **130** to a circulator **806**. It should be noted that first fiber optic cable **804** may be a single fiber optic cable or a plurality of fiber optic cables connected together. Additionally, first fiber optic cable **804**, as well as any fiber optic cable disclosed herein, may comprise a single optical fiber or a plurality of optical fibers. A plurality of optical fibers may allow for multi-fiber sensing and/or multimode measurement operations. Circulators are passive optical devices. Passive optical devices are devices that are optically connected to a fiber optic cable to receive and direct light (e.g., light pulses) along the fiber optic cable or to at least another fiber optic cable connected to the passive optical device. Although circulators are described herein for this disclosure, any suitable passive optical device may be utilized in place of the circulator. Examples of suitable passive optical devices comprise a fused type fiber optic splitter, a Planar Waveguide Circuit (PLC) fiber optic splitter, or any other type of optical splitter. Circulator **806** may connect first fiber optic cable **804** to second fiber optic cable **808**, which may be referred to as an upgoing fiber. It should be noted that second fiber optic cable **808** may be single fiber optic cable or a plurality of fiber optic cables connected together. Additionally, second fiber optic cable **808**, as well as any fiber optic cable disclosed herein, may comprise a single optical fiber or a plurality of optical fibers. A plurality of optical fibers may allow for multi-fiber sensing and/or multimode measurement operations. In examples, circulator **806** functions to steer light unidirectionally between one or more input and outputs of circulator **806**. Without limitation, circulators **806** are passive three-port devices wherein light from a first port is split internally into two independent polarization states and wherein these two polarization states are made to propagate two different paths inside circulator **806**. These two independent paths allow one or both independent light beams to be rotated in polarization state via the Faraday effect in optical media. Polarization rotation of the light propagating through free space optical elements within the circulator thus allows the total optical power of the two independent beams to uniquely emerge together with the same phase relationship from a second port of circulator **806**.

[0069] Conversely, if any light enters the second port of remote circulator **806** in the reverse direction, the internal free space optical elements within remote circulator **806** may operate identically on the reverse direction light to split it into two polarizations states. After appropriate rotation of polarization states, these reverse in direction polarized light beams, are recombined, as in the forward propagation case, and emerge uniquely from a third port of remote circulator **806** with the same phase relationship and optical power as they had before entering remote circulator **806**. Additionally, as discussed below, remote circulator **806** may act as a gateway, which may only allow chosen wavelengths of light to pass through remote circulator **806** and pass to downhole sensing fiber **132**. Second fiber optic cable **808** may attach umbilical line **130** to input **810**. Input **810** may be a fiber optic connector which may allow backscatter light to pass into interrogator unit **128** to interferometer **706** Interferometer **706** may operate and function as described above and further pass back scatter light to photodetector assembly **718**.

[0070] FIG. **9** illustrates another example of FOS system **126**, which is a DAS system. As illustrated, interrogator unit **128** may comprise one or more DAS interrogator units **900**, each emitting coherent light pulses at a distinct optical wavelength, and a Raman Pump **902** connected to a wavelength division multiplexer **904** (WDM) with fiber stretcher (s). In examples, any type of optical amplifier may be utilized in place of Raman Pump **902**. Without limitation, WDM **904** may comprise a multiplexer assembly that multiplexes the light received from the one or more DAS interrogator units **900** and a Raman Pump **902** onto a single optical fiber and a demultiplexer assembly that separates the multi-wavelength backscattered light into its individual frequency components and redirects each single wavelength backscattered light stream back to the corresponding DAS interrogator unit **900**. However, although WDM's are described above and below, WDM's may be substituted for with a diffraction grating filter (DGF), a holographic diffraction grating filter, an optical Fourier filter (i.e., tunable or static), a nonlinear frequency-division multiplexer (NFDM), an arrayed waveguide grating (AWG), and/or a quantum-memory wavelength-division multiplexer (QWDM). Additionally, it should be noted that WDMs may operate and function to combine and/or split light (e.g., signals) that pass through the WDM or equivalent. Furthermore, WDMs may enable time-division multiplexed (TDM) or frequency-divisional multiplexed (FDM) interrogation systems.

[0071] In an example, WDM **904** may utilize an optical add-drop multiplexer to enable multiplexing the light received from the one or more DAS interrogator units **900** and a Raman Pump **902** and demultiplexing the multi-wavelength backscattered light received from a single fiber. WDM **904** may also comprise circuitry to optically amplify the multi-frequency light prior to launching it into the single optical fiber and/or optical circuitry to optically amplify the multi-frequency backscattered light returning from the single optical fiber, thereby compensating for optical losses introduced during optical (de-) multiplexing. Raman Pump **902** may be a co- and/or counter-propagating optical pump based on stimulated Raman scattering, to feed energy from a pump signal to a main pulse from one or more DAS interrogator units **900** as the main pulse propagates down one or more fiber optic cables. This may conservatively yield a distributed 3 dB improvement in SNR at 25 km tie-back distances, but as much as a distributed 6 dB gain at 50 km. Moreover, Raman Pump **902** may comprise of cascading Raman amplifiers assembled to yield a distributed gain profile along the transmission fiber(s).

[0072] As illustrated, Raman Pump **902** is located in interrogator unit **128** for co-propagation. In another example, Raman Pump **902** may be located topside after one or more circulators **806** either in line with first fiber optic cable **804** (co-propagation mode) and/or in line with second fiber optic cable **808** (counter-propagation). In another example, Raman Pump **902** is marinized and located after distal circulator **908** configured either for co-propagation or counter-propagation. In still another example, the light emitted by the Raman Pump **902** is remotely reflected by using a wavelength-selective filter beyond a circulator **806** in order to provide amplification in the return path using a Raman Pump **902** in any of the topside configurations outlined above. The wavelength-selective filter beyond circulator **806** may also be used to ensure the high optical power of the Raman Pump **902** is reflected from low power optical components, such as the optical wet-mate connectors in the optical feedthrough system **144** (e.g., referring to FIGS. **6**A and **6**B).

[0073] Further illustrated in FIG. **9**, WDM **904** with fiber stretcher may attach proximal circulator **906** to umbilical line **130**. Umbilical line **130** may include one or more remote circulators **806**, a first fiber optic cable **804**, and a second fiber optic cable **808**. As illustrated, a first fiber optic cable **804** and a second fiber optic cable **808** may be separate and individual fiber optic cables that may be attached at each end to one or more remote circulators **806**. In examples, first fiber optic cable **804** and second fiber optic cable **808** may be different lengths or the same length and each may be an ultra-low loss transmission fiber that may have a higher power handling capability before non-literarily. This may enable a higher gain, co-propagation Raman amplification from interrogator unit **128**.

[0074] Deploying first fiber optic cable **804** and as second fiber optic cable **808** from floating vessel **102** (e.g., referring to FIGS. **1**A and **1**B) to a subsea environment to a distal-end passive optical circulator arrangement, enables downhole sensing fiber **132**, which is a sensing fiber, to be below a remote circulator **806** (e.g., well-only) that may be at the distal end of FOS system **126**, which is a DAS system. This may allow for higher (e.g., 2-3×) DAS pulse repetition rates, and allow for the optical receivers to be adjusted such that their dynamic range is optimized for downhole sensing fiber **132**. Depending on the tie-back distance between OFL and interrogator, this may yield 3 to 10 dB improvement in SNR. Additionally, downhole sensing fiber **132** may be an enhanced backscatter sensing fiber that has higher-than-Rayleigh scattering coefficient which may result in a ten to one hundred times improvement in backscatter, which may yield a 10 dB to 20 dB improvement in SNR. In examples, remote circulators **806** may further be categorized as a proximal circulator **906** and a distal circulator **908**. Proximal circulator **906** is located closer to interrogator unit **128** and may be located on floating vessel **102** or within umbilical line **130**. Distal circulator **908** may be further away from interrogator unit **128** than proximal circulator **906** and may be located in umbilical line **130**, in an optical flying lead **142**, in an optical feedthrough system **144**, or within wellbore **122** (e.g., referring to FIGS. **1**A and **1**B). As discussed above, a configuration illustrated in FIG. **8** may not utilize a proximal circulator **906** with lead lines **800**.

[0075] FIG. **10** illustrates another example of distal circulator **908**, which may include two remote circulators **806**. As illustrated, each remote circulator **806** may function and operate to avoid overlap, at interrogator unit **128**, of backscattered light from two different pulses. For example, during operations, light at a first wavelength may travel from interrogator unit **128** down first fiber optic cable **804** to a remote circulator **806**. As the light passes through remote circulator **806** the light may encounter a Fiber Bragg Grating **1000**. In examples, Fiber Bragg Grating **1000** may be referred to as a filter mirror that may be a wavelength specific high reflectivity filter mirror or filter reflector that may operate and function to recirculate unused light back through the optical circuit for "double-pass" co- and/or counter-propagating Raman amplification of the DAS signal. In examples, Fiber Bragg Grating **1000** may be referred to as an optically reflective element. In examples, this wavelength specific "Raman light" mirror may be a dichroic thin film interference filter, Fiber Bragg Grating **1000**, or any other suitable optical filter that passes only the 1550 nm forward propagating DAS interrogation pulse light while simultaneously reflecting most of the residual Raman Pump light.

[0076] Without limitation, Fiber Bragg Grating **1000** may be set-up, fabricated, altered, and/or the like to allow only certain selected wavelengths of light to pass. All other wavelengths may be reflected back to the second remote circulator, which may send the reflected wavelengths of light along second fiber optic cable **808** back to interrogator unit **128**. This may allow Fiber Bragg Grating **1000** to split DAS system **126** (e.g., referring to FIG. **9**) into two regions. A first region may be identified as the devices and components before Fiber Bragg Grating **1000** and the second region may be identified as downhole sensing fiber **132** and any other devices after Fiber Bragg Grating **1000**.

[0077] Splitting the DAS system, which is a FOS system **126**, (e.g., referring to FIG. **9**) into two separate regions may allow interrogator unit **128** (e.g., referring to FIGS. **1**A and **1**B) to pump specifically for an identified region. For example, the disclosed system of FIG. **9** may include one or more Raman pumps **902**, as described above, placed in interrogator unit **128** or after proximal circulator **906** at the topside either in line with first fiber optic cable **804** or second fiber optic cable **808** that may emit a wavelength of light that may travel only to a first region and be reflected by Fiber Bragg Grating **1000**. A second Raman pump may emit a wavelength of light that may travel to the second region by passing through Fiber Bragg Grating **1000**. Additionally, both the first Raman pump and second Raman pump may transmit at the same time. Without limitation, there may be any number of Raman pumps and any number of Fiber Bragg Gratings **1000** which may be used to control what wavelength of light travels through downhole sensing fiber **132**. FIG. **10** also illustrates Fiber Bragg Gratings **1000** operating in conjunction with any remote circulator **806**, whether it is a distal circulator **908** or a proximal circulator **906**. Additionally, as discussed below, Fiber Bragg Gratings **1000** may be attached at the distal end of downhole fiber **218** and act as a mirror. Other alterations to DAS system **126** (e.g., referring to FIG. **9**) may be undertaken to improve the overall performance of DAS system **126**. For example, the lengths of first fiber optic cable **804** and second fiber optic cable **808** may be selected to increase pulse repetition rate (expressed in terms of the time interval between pulses t_{rep}).

[0078] FIG. **11** illustrates an example of fiber optic cable **1100** in which no remote circulator **806** may be used. As illustrated, the entire fiber optic cable **1100** is a sensor and the pulse interval may be greater than the time for the pulse of light to travel to the end of fiber optic cable **1100** and its backscatter to travel back to interrogator unit **128** (e.g., referring to FIGS. **1**A and **1**B). This is so, since in the DAS system, which is a FOS system **126**, at no point in time, backscatter from more than one location along sensing fiber (i.e., downhole sensing fiber **132**) may be received. Therefore, the pulse interval t_{rep} may be greater than twice the time light takes to travel "one-way" down the fiber. Let t_s be the "two-way" time for light to travel to the end of fiber optic cable **1100** and back, which may be written as $t_{rep}>t_s$.

[0079] FIG. **12** illustrates an example of fiber optic cable **1100** with a remote circulator **806** using the configuration shown in FIG. **9**. When a remote circulator **806** is used, only the light traveling in fiber optic cable **600** that is allowed to go beyond remote circulator **806** and to downhole sensing fiber **132** may be returned to interrogator unit **128** (e.g., referring to FIGS. **1**A and **1**B), thus, the interval between pulses is dictated only by the length of the sensing portion, downhole sensing fiber **132** of fiber optic cable **1100**. It should be noted that in terms of pulse timing what matters is the two-way travel time of the light pulse "to" and "from" the sensing portion, downhole sensing fiber **132**. Therefore, the first fiber optic cable **804** or second fiber optic cable **808** "to" and "from" remote circulator **806** may be longer than the other, as discussed above.

[0080] FIG. **13** illustrates an example remote circulator arrangement **1300** which may allow, as described above, configurations that use more than one remote circulator **806** close together at the remote location. Although remote circulator arrangement **1300** may have any number of remote circulators **806**, remote circulator arrangement **1300** may be illustrated as a single remote circulator **806**.

[0081] FIG. **14** illustrates an example first fiber optic cable **804** and second fiber optic cable **808** attached to a remote circulator **806** at each end. As discussed above, each remote circulator may be categorized as a proximal circulator **906** and a distal circulator **908**. When using a proximal circulator **906** and a distal circulator **908**, light from the fiber section before proximal circulator **906**, and light from the fiber section below the remote circular **806** are detected, which is illustrated in FIGS. **15**A-**16**. There is a gap **1500** between them of "no light" that depends on the total length of fiber (summed) between proximal circulator **906** and a distal circulator **908** (e.g., referring to FIG. **14**).

[0082] Referring back to FIG. **14**, with t_{s1} the duration of the light from fiber sensing section before proximal circulator **906**, t_{sep} the "dead time" separating the two sections (and due to the cumulative length of first fiber optic cable **804** and second fiber optic cable **808** between proximal circulator **906** and a distal circulator **908**), and t_{s2} the duration of the light from the sensing fiber, downhole sensing fiber **132**, beyond distal circulator **908**, the constraints on fiber lengths and pulse intervals may be identified as:

i

$$t_{rep} < t_{sep}. \tag{3}$$

ii

$$(2t_{rep}) > (t_{s1} + t_{sep} + t_{s2}). \tag{4}$$

Criterion (i) ensures that "pulse n" light from downhole sensing fiber **132** does not appear while "pulse n+1" light from fiber before proximal circulator **906** is being received at interrogator unit **128** (e.g., referring to FIGS. **1**A and **1**B). Criterion (ii) ensures that "pulse n" light from downhole sensing fiber **132** is fully received before "pulse n+2" light from fiber before proximal circulator **906** is being received at interrogator unit **128**. It should be noted that the two criteria given above only define the minimum and maximum t_{rep} for scenarios where two pulses are launched in the fiber before backscattered light below the remote circulator **806** is received. However, it should be appreciated that for those skilled in the art these criteria may be generalized to cases where n E {1, 2, 3, . . . } light pulses may be launched in the fiber before backscattered light below the remote circulator **806** is received.

[0083] The use of remote circulators **806** may allow for FOS system **126**, a DAS system, (e.g., referring to FIG. **8**) to increase the DAS pulse repetition rate, or sampling frequency. FIG. **17** illustrates workflow **1700** for optimizing sampling frequency when using a remote circulator **806** in FOS system **126**. One skilled in the art will appreciate the subtlety that optimizing the sampling frequency doesn't imply maximizing the sampling frequency. Workflow **1700** may begin with block **1702**, which determines the overall fiber length in both directions. For example, in case of a 17 km of first fiber optic cable **804** and 17 km of second fiber optic cable **808** before distal circulator **908** and 8 km of sensing fiber, downhole sensing fiber **132**, after distal circulator **908**, the overall fiber optic cable length in both directions would be 50 km. Assuming a travel time of the light of 5 ns/m, the following equation may be used to calculate a first DAS sampling frequency f_s

$$f_s = \frac{1}{t_s} = \frac{1}{5 \cdot 10^{-9} \cdot z} \tag{5}$$

where t_s is the DAS sampling interval and z is the overall two-way fiber length. Thus, for an overall two-way fiber length of 50 km the first DAS sampling rate f_s is 4 kHz. In block **1704** regions of the fiber optic cable are identified for which backscatter is received. For example, this is done by calculating the average optical backscattered energy for each sampling location followed by a simple thresholding scheme. The result of this step is shown in FIG. **15**A where boundaries **1502** identify two sensing regions **1504**. As illustrated in FIG. **15**A-**15**C, optical energy is given as:

$$I^2 + Q^2 \tag{6}$$

where I and Q correspond to the in-phase (I) and quadrature (Q) components of the backscattered light. In block **1706**, the sampling frequency of FOS system **126**, a DAS system, is optimized. To optimize the sampling frequency a minimum time interval is found that is between the emission of light pulses such that at no point in time backscattered light arrives back at interrogator unit **128** (e.g., referring to FIG. **1**) that corresponds to more than one spatial location along a sensing portion of the fiber-optic cable. Mathematically, this may be defined as follows. Let S be the set of all spatial sample locations x along the fiber for which backscattered light is received. The desired light pulse emission interval t_s is the smallest one for which the cardinality of the two sets S and $\{\text{mod}(x,t_s):x\in S\}$ is still identical, which is expressed as:

$$\min_{t_s}(t_s) \text{ s.t. } |S| = |\{\text{mod}(x, t_s) : x \in S\}| \tag{7}$$

where |•| is the cardinality operator, measuring the number of elements in a set. FIG. **16** shows the result of optimizing the sampling frequency from FIGS. **10**A-**10**C with workflow **1700**. Here, the DAS sampling frequency may increase from 4 kHz to 12.5 kHz without causing any overlap in backscattered locations, effectively increasing the signal to noise ratio of the underlying acoustic data by more than 5 dB due to the increase in sampling frequency.

[0084] Variants of FOS system **126**, which may be DAS based, may also benefit from workflow **1700**. For example, FIG. **18** illustrates FOS system **126** in which proximal circulator **906** is placed within interrogator unit **128**. This system set up of FOS system **126** may allow for system flexibility on how to implement during measurement operations and the efficient placement of Raman Pump **1900**. As illustrated in FIGS. **19** and **20**, first fiber optic cable **804** and second fiber optic cable **808** may connect interrogator unit **128** to umbilical line **130**, which is described in greater detail above in FIG. **8**.

[0085] FIG. **19** illustrates another example of DAS system **126** in which Raman Pump **1900** is operated in co-propagation mode and is attached to first fiber optic cable **804** after proximal circulator **906**. For example, if the first sensing region before proximal circulator **906** should not be affected by Raman amplification. Moreover, Raman Pump **1900**, may also be attached to second fiber optic cable **808** which may allow the Raman Pump **1900** to be operated in counter-propagation mode. In examples, the Raman Pump may also be attached to fiber **1902** between WDM **904** and proximal circulator **906** in interrogator unit **128**.

[0086] FIG. **20** illustrates another example of FOS system **126** in which an optical amplifier assembly **2000** (i.e., an Erbium doped fiber amplifier (EDFA)+Fabry-Perot filter) may be attached to proximal circulator **906**, which may also be identified as a proximal locally pumped optical amplifier. In examples, a distal optical amplifier assembly **2002** may also be attached at distal circulator **908** on first fiber optical cable **804** or second fiber optical cable **808** as an inline or "mid-span" amplifier. In examples, optical amplifier assembly **2002** located in-line with fiber optical cable **804** and above distal circulator **908** may be used to boost the light pulse before it is launched into the downhole sensing fiber **132**. Referring to FIGS. **10**B and **10**C, the effect of using an optical amplifier assembly **2000** in-line with a second fiber optic cable **808** prior to proximal circulator **906** and/or using an distal optical amplifier assembly **2002** located in line with second fiber optical cable **808** above distal circulator **908** may allow for selectively amplifying the backscattered light originating from downhole sensing fiber **132** which tends to suffer from much stronger attenuation as it travels back along downhole sensing fiber **132** and second fiber optical cable **808** than backscattered light originating from shallower sections of fiber optic cable that may also perform sensing functions. FIG. **10**B illustrates measurements where proximal circulator **906** is active (optical amplifier assembly **2000** in-line with a second fiber optic cable **808** prior to proximal circulator **906** and/or distal optical amplifier assembly **2002** located in line with second fiber optical cable **808** above distal circulator **908** is used). FIG. **10**C illustrates measurements where proximal circulator **906** is passive (no optical amplification is used in-line with second fiber optic cable **808**). In FIGS. **10**B and **10**C, boundaries **1502** identify two sensing regions **1504**. Additionally, in FIGS. **10**B and **10**C the DAS sampling frequency is set to 12.5 kHz using workflow **1700**. Further illustrated Fiber Bragg Grating **1000** may also be disposed on first fiber optical cable **804** between distal optical amplifier assembly **2002** and distal circulator **908**.

[0087] During operation, data quality from a FOS system **126**, such as a DAS system, (e.g., referring to FIG. **7**) may be governed by signal quality and sampling rate. Signal quality is predominantly constrained by the power of back-scattered light and sampling rate is constrained by sensing fiber length. For example, the less backscattered light that is received from a sensing fiber, which may be downhole sensing fiber **132** or disposed on downhole sensing fiber **132** (e.g., referring to FIGS. **1**A and **1**B), the more inferior the quality of the measurement taken by FOS system **126**.

[0088] FIG. **21** illustrates FOS system **126** in which one or more self-powered optical amplifiers (SPOA) **2100** are disposed at one or more locations within FOS system **126**. As discussed above, FOS system **126** is a fiber optical system, thus a fiber optic line or a plurality of fiber optic lines may be utilized to connect each SPOAs **2100** to each other. By utilizing SPOAs **2100**, the reach of FOS system **126** may be increased over the current reach of traditional fiber optic systems. As illustrated, fiber optic cables may traverse for up to twenty kilometers between each SPOA **2100**, and in some instances much longer up to 100 km+depending on the subsea infrastructure. Thus, interrogation signal **2102**, which may be created from a continuous light wave or pulsed, originating from interrogator **128** may be amplified during operations to increase the distance between transmission of interrogation signal **2102** and use of interrogation signal **2102** to perform measurement operations with downhole sensing fiber **132**. With continued reference to FIG. **21**, an SPOA **2100** may be disposed between downhole sensing fiber **132** and distal circulator **908** to amplify both the forward interrogation signal and backward scattered signal. A backward scatter signal is formed from interrogation signal **2102** reflecting off the bottom of downhole sensing fiber **132**, which may travers back to interrogator **129** by way of distal circulator **908** and proximal circulator **906** using second fiber optic cable **808**. As noted above, SPOAs **2100** may be disposed on both first fiber optic cable **804** and second fiber optic cable **808**.

[0089] FIG. **22** illustrates an example of SPOA **2100**. It should be noted that the components of SPOA **2100** described herein are used for FIGS. **23-25**. As illustrated, interrogation signal **2102** may enter SPOA **2100** utilizing first fiber optic cable **804**. In other examples, interrogation signal **2102** may enter SPOA **2100** using downhole sensing fiber **132**. Additionally, a backscatter signal may enter SPOA **2100** utilizing second fiber optic cable **808**. As noted above, all fiber optic cables (i.e., first fiber optic cable **804**, second fiber optic cable **808**, downhole sensing fiber **132**) may comprise a single optical fiber or a plurality of optical fibers. In examples, interrogation signal **2102** may enter SPOA **2100** through proximal tap **2200**. A proximal tap **2200** may be used to split a small fraction (e.g., 1%) of interrogation signal **2102** or a fraction of interrogation signal **2102** in the range of 0.1% to 10% of interrogation signal **2102** where the small fraction of the signal is used to i) monitor amplitude(s) of the interrogator wavelengths ii) monitor for any control signal iii) measure any back scatter signals from the gain element. A WDM (Wavelength Division Multiplexer) or a CWDM (Coarse Wavelength Division Multiplexer) may be used to de-multiplex a control signal in the L-band wavelength range from the Interrogator lasers in the C-band wavelength range. Proximal tap **2200** may be a micro-optic device or a fused fiber device or in some instances a combination of a fused fiber tap coupler followed by a micro-optic device allowing separation of the different signal wavelengths and monitoring of the individual wavelengths in the system. Further, proximal tap **2200** may be selected to measure signals propagation from the surface interrogator towards the distal circulator and/or measure signals like ASE generated in SPOA **2100** and/or other back scattered signals. During operations, proximal tap **220** may operate and function to measure the amplitude of the transmitted signals and this information can then be used to set pump laser operating points to target a specific signal gain in Erbium Doped fiber **2210**. Further, proximal tap **2200** may comprise an Amplified Stimulated Emission (ASE) filter **2202**. ASE filter **2202** may operate and function to filter out any broad band ASE (amplified Spontaneous Emission) power at wavelengths outside the laser signals as ASE may generate noise in amplified systems. In examples, ASE filter **2202** blocks energy outside the signal wavelengths of interrogation signal **2102**. Erbium doped amplifiers have a broad band gain window and some of the gain outside the laser signals appear as noise that be suppressed. For example, small signal gain is in general higher in amplified systems and any out of band small signals should be suppressed in order to limit ASE noise from reaching following amplification stages as it could then be amplified in subsequent amplifiers with associated increase in noise floor and/or reduction in target signal gain as some of the available gain in the amplifier would be used to amplify noise instead of amplifying actual signals. Out of band signals may also increase non-linear signal distortion along the fiber span, such as Erbium doped fiber **2210**, between SPOAs **2100** if allowed to interact uncontrolled within the system. As interrogation signal **2102** may traverse through proximal tap **2200** and/or ASE filter **2202**, each interrogation signal **2102** may be transmitted to a proximal WDM **2204** by communication optical fiber **2206**. It should be noted that proximal WDM **2204** may operate and function as WDM **904**, described above in FIG. **9**.

[0090] Communication optical fiber **2206** is a standard fiber optic cable utilized to communicate interrogation signal **2102** between different components within SPOA **2100**. Additionally, communication optical fiber **2206**, as with all fiber optic cable disclosed here, may comprise a single optical fiber or a plurality of optical fibers. As illustrated, interrogation signal **2102** may be input in to proximal WDM **2204**, a reflector **2208** may be at least a part of the proximal WDM **2204** and/or connected to proximal WDM **2204** by a communication optical fiber **2206**. Reflector **2208** may reflect light from pump **2214**, discussed below, back into Erbium doped fiber **2210**. This may allow WDM **2204** to direct the light from pump **2214** away from/reduce any light from pump **2214** from hitting the tap **2200** as it may distort signal readings and potentially force the use of WDMs **2204** and/or ASE filters **2202** on interrogation signal **2102**. An alternative implementation would be to use a low reflectance termination for device **2208**.

[0091] Interrogation signal **2102** may exit proximal WDM **2204** and enter Erbium doped fiber **2210** (also may be referred to as Erbium doped fiber amplifiers), which may be connected to proximal WDM **2204**. Erbium doped fiber **2210** may be formed from adding Erbium ions into an optical fiber during the fiber manufacturing process. Erbium doped fiber amplifiers may allow for amplification of interrogation signals **2102** operating around C-band when the Erbium fiber is pumped by the 980 nm band and/or 1480 nm bands laser light traversing through Erbium doped fiber **2210**. Additionally, the photons from light emitted by pump **2214** may excite Erbium ions to a higher energy state within the Erbium doped fiber **2210** and any signals in the 1550 nm C-band may then be amplified as the ions give up their energy to the signal and return to their lower energy state. Erbium doped fiber amplifiers may operate in the C-band which may be around 1525 nm-1565 nm and it is desirable to pump with 980 nm lasers if/when sufficient gain can may be achieved given that the 980 nm pumped Erbium doped fiber amplifiers tend to have lower noise gain performance due to the higher pump absorption cross section. In examples, a 1480 nm pump band may have a lower and broader absorption cross section and is generally used for high power amplifiers. A combination of 980 nm and 1480 nm lasers may be used in Erbium amplifiers depending on the gain parameters. Small signal gain amplifiers may be preferentially pumped with 980 nm lasers to increase signal to noise and large signal gain may be pumped with a combination of 980 nm and 1480 nm pump **2214**. Optical fibers may be doped with other materials to achieve gain in other wavelengths where (e.g., Ytterbium may be added) to add gain in the L-band 1565 nm-1610 nm or Thulium may be added to add gain in the S-band 1450 nm-1490 nm or Praseodymium amplifiers may operate in the 1300 nm region. Additionally, Erbium doped fiber **2210** may operate and function to generate gain along the length of Erbium doped fiber **2210**. As illustrated in FIGS. **26**A-**26**E, cladding pumped fiber amplifiers **2600** that may be coupled and combine multiple shortwave Raman pump lasers onto a common LEAF single mode transmission fiber to generate gain along the length of Erbium doped fiber **2210** (e.g., referring to FIGS. **22**-**25**. FIGS. **26**B-**26**E are different cross sections of Erbium doped fiber **2210**. Use large area fibers, (e.g., 105 um MMF) for pump **2214**/**2302** in order to lower optical power density across optical connectors (unless spliced) and then use an MMF/SMF cladding pumped amplifiers supporting gain at 1550 nm (1.55 um) or other doped earth amplifiers to generate gain in the signal bands of interest. Interrogation signal **2102** traversing through Erbium doped fiber **2210** may enter distal WDM **2212**. It should be noted that proximal WDM **2204** may operate and function as WDM **904**, described above in FIG. **9**. At distal WDM **2212**, interrogation signal **2102** may be amplified by energy originating from pump **2214**. In examples, energy may be combined with interrogation signal **2102** within distal WDM **2212**.

[0092] Pump **2214** may be 980 nm pump that may be used for low noise applications for small signal gains on the return path to the surface for backscattered light. In examples, a combination of both 980 nm and 1480 nm band pumps may be utilized for the downward travelling transmitted pulse. The 1480 nm pump may be used for remote pumping as the fiber attenuation is less at 1480 nm for use in remote amplification whereas a 980 nm pump may experience excessive attenuation and not be usable if placed a large distance (km's) from Erbium doped fiber **2210**. As illustrated pump **2214** may connect to distal WDM **2212** by one or more communication optical fibers **2206** and an optional circulator **2216** disposed within SPOA **2100**. Circulator **2216** may operate and function as described above regarding any circulator. The energy transmitted from pump **2214** may be controlled by controller **2218**. As illustrated, controller **2218** may be connected to pump **2214** by communication wire **2220**. Communication wire **2220** may be a standard composite metal wire or a fiber optic cable that may comprise one or more optical fibers. This may allow of control signals that may be electrical or optical in nature. As illustrated, communication wire **2220** may allow for control signals to operate and function pump **2214**. To properly control the energy transmitted from pump **2214**, controller **2218** may be connected to proximal tap **2200** and distal tap **2222** by one or more communication optical fibers **2206**. Further, controller **2218** is connected to circulator **2216** by a communication optical fiber **2206**. It should be noted that all optical fibers may connect to controller **2218** through an optical-to-electrical (O/E) conversion connection. This may allow for controller **2218** to sample interrogation signal **2102** that may be entering proximal tap **2200** and interrogation signal **2102** as it leaves distal tap **2222** on fiber optic line **804** after being amplified by energy from pump **2214**. Additionally, controller **2218** may sample from the energy being transmitted from pump **2214** through circulator **2216** as well as sample the energy directly from power supply **2224**, to be described below, by a communication wire **2220**, which feeds pump **2214** through another communication wire **2220**. In other examples, pump **2214** may be monitored by selecting a pump that may comprise a back facet monitoring diode where the optical power may be monitored and the electrical signal from the back facet monitor may be electrically connected to the controller and thereby monitored. The optical power output from a pump laser is in general proportional to the laser drive current once the lasing threshold has been exceeded at the beginning of the laser service life, and a simple control scheme may use a calibration table of pump laser output vs. laser drive current for a given pump. The laser drive current vs. laser optical output power relationship may change over time as component age and it may be beneficial to have means for closed loop control of the optical pump power using the back facet monitor or an optical pump laser power monitoring signal.

[0093] Controller **2218** may operate and function to maintain optimal gain condition. This may be performed by measuring input and output signal levels through taps **2200**, **2222** and measure the signal gain through SPOA **2100**. To achieve a fixed gain, or the desired output signal level may be pre-determined prior to SPOA **2100** deployment, or the gain setpoint may be selected and adjusted by a supervisory system locally in SPOA **2100** by monitoring signal levels/ASE levels/back reflection from connectors or by measuring signal levels/pump laser settings/ASE levels/back reflection from connectors. The information may be communicated using an optical communication channel to a supervisory system on surface **103** (e.g., referring to FIGS. **1**A and **1**B) where the supervisory system controls each SPOA **2100** and incrementally and/or iteratively control interrogator laser settings, surface amplifier settings on the transmission path, SPOAs **2100** from surface **103** to sensing fiber **132**, SPOAs **2100** from sensing fiber **132** to interrogators **129**, surface amplifier settings on the return path and receiver opto-electric settings. The system settings may be controlled as function of signal to noise, system reliability as it is desired to operate pumps **2213**, **2302** within a target range for long term reliability, avoid connector damage and excessive power levels within any system component and balance system signal levels as interrogator settings may be changed for various sensing applications where pulse width and gauge length may be changed to optimize SNR for various applications. This may allow controller **2218** to amplify interrogation signal **2102** as they pass through SPOA **2100**.

[0094] FIG. **23** illustrates a SPOA **2100** that may further comprise a 2×2 switch **2300**. In this example, 2×2 switch **2300** may allow for interrogation signal **2102** to traverse through 2×2 switch **2300** and bypass amplification from SPOA **2100**. This may allow for interrogation signal **2102** to continue to traverse through first fiber optic cable **804**, second fiber optic cable **808**, and/or downhole sensing fiber **132** if SPOA **2100** is not operating and/or functioning. For example, during operation, 2×2 switch **2300** may be in cross-path setting so that a path through SPOA **2100** is included. This setting may be controlled by controller **2218**, which may be connected to 2×2 switch through communication wire **2220**. If a failure is detected within SPOA **2100**, 2×2 switch **2300** may be set to blocked path position in which case the signal paths on the left side is directly connected to the signal path on the right, thereby bypassing the bottom portion of SPOA **2100** where the optical amplification takes place. Referring back to FIG. **22**, if SPOA **2100** is not operating and/or functioning, then interrogation signal **2102** may not be able to bypass SPOA **2100**. Interrogation signal **2102** may then dissipate within SPOA **2100** and never leave. The default position for optical switch **2300** is the low loss path and the switch may mechanically set to the default position in case of power loss.

[0095] Referring back to FIG. **23**, SPOA **2100** may further comprise a second pump **2302**. Second pump **2302** may be utilized if first pump **2214** fails. In this example, a 1×2 switch **2304** may be connected to first pump **2214** and second pump **2302** by communication optical fiber **2206**. A 1×2 switch **2304** may route all the wavelengths from one fiber to another fiber. Likewise, 1×2 switch **2304** may be connected to distal WDM **2212** by communication optical fiber **2206**. In other examples, first pump **2214** and second pump **2302** may have two different wavelengths (1450 nm vs 1460 nm, for example) and be combined using a WDM (Wavelength Division Multiplexer) or a polarization multiplexer. A CWDM (Coarse Wavelength Division Multiplexer) may be used to de-multiplex a control signal in the L-band wavelength range from the Interrogator lasers in the C-band wavelength range. The function would be similar to an Optical Add Drop Multiplexer (OADM) where specific wavelengths may be routed but with different signal attenuation characteristics. A CWDM may in some instances be a wide band fused fiber-based device with low loss whereas a OADM is highly likely a micro-optic device with higher loss and narrow signal bands. A CWDM or a OADM may be used in **2200** and/or **2222** to de-multiplex a control signal that is used to communicate with controller **2218**. If first pump **2214** fails, second pump may be turned on by controller **2218** and/or 1×2 switch **2304** may be used to switch from first pump **2214** to second pump **2302** using controller **2218**, which may connect to 1×2 switch **2304** by communication wire. The light emitted from pumps **2214**, **2302** may also be selected at around 980 nm and around 1480 nm in order to optimize noise and signal gain. Alternatively, multiple pumps at the same wavelength or similar wavelengths may be used and a polarization division multiplexer in place of 1×2 switch **2304** where both pumps **2214**, **2302** are operated simultaneously at a lower operating point thus improving reliability and operating life of the individual components. Redundance may be further added to SPOA **2100** using different configurations of switches.

[0096] FIG. **24** illustrates a SPOA **2100** in which multiple components may be utilized for redundance. These multiple components may be accessed using a proximal 1×N switch **2400** and a distal 1×N switch **2402**. The use of proximal 1×N switch **2400** and distal 1×N switch **2402** may also allow for a bypass to optical amplification using bypass fiber optic cable **2406**. Bypass fiber optic cable **2406** is the default state if the power source **2224** fails to provide power. The optical switches and any signal routing elements would default to a low loss signal path through SPOA **2100** to provide signal integrity if there is a failure of power source **2224**. A system with multiple SPOAs **2100** may be used where the optical gain characteristics for FOS system **126** as a whole may be maintained if a single SPOA **2100** fails for any reason where the failure is loss of power, fiber breakage along the gain path or other events that may cause SPOA **2100** to not perform as planned. Bypass fiber optic cable **2406** may be enabled by loss of power where the switch default state is controlled using a mechanical configuration with spring loaded signal actuators, by controller **2218** if it cannot achieve specific gain conditions, by controller **2218** if SPOA **2100** cannot be achieved due to component failure or other similar reasons. As noted above, all fiber optic cables may comprise a single optical fiber or a plurality of optical fibers. Both proximal 1×N switch **2400** and distal 1×N switch **2402** may be controlled and/or operated by controller **2218**, which is connected to proximal 1×N switch **2400** and distal 1×N switch **2402** by a communication wire **2220**. This may allow controller **2218** to select a channel in proximal 1×N switch **2400** and the same channel in distal 1×N switch **2402**. Controller **2218** may monitor available power from power supply **2224** and communicate power levels to information handling system **146** thus enabling information handling system **146** to control individual SPOA optical gain and power draw while managing the optical system signal to noise. This allows FOS system **126** to manage available electrical power in order to maintain system performance during exceptions, start up scenarios where power source may not be fully operational of end of life situations of power sources.

[0097] In FIGS. **22-25**, controller **2218** may have a Tx/Rx transceiver port **2500** so that controller **2218** may communicate with information handling system **146** at surface **103** (e.g., referring to FIGS. **1**A and **1**B) through communication wire **2220**. While FIGS. **22-25** only show a single fiber going in and out of SPOA **2100**, there may be a plurality of fibers from first fiber optic cable **804**, downhole sensing fiber **132**, or second fiber optic cable **808** to enable multi-fiber sensing and/or enable future expansion as additional wellbores **122** (e.g., referring to FIGS. **1**A and **1**B) may be added over time. This may allow for some fibers within first fiber optic cable **804** or second fiber optic cable **808** to be terminated in subsea ROV mateable subsea connectors to enable communication and/or sensing system expansions as needed to tie in additional wellbores **122**, fields or enable

ROV connections to bypass one or more SPOAs **2100**. Similarly, there may be multiple bypass fiber optic cable **2406** as well as multiple amplification paths, as discussed above in FIGS. **22-25**. Additionally, a communication signal **2502**, which may be wired and/or wireless, to information handling system **146** may be made through the existing fiber optic cable. As illustrated Tx/Rx transceiver port **2500** may transmit or receive communication signal **2502** through an Optical Add Drop Multiplexer (OADM) **2504**. As illustrated in FIG. **25** a dedicated International Telecommunication Union (ITU) channel may be used and OADM **2504** may be added to combine/split the communication link at the SPOA location. An example optical arrangement is shown below (communication signal is indicated by dashed line). Communication signal **2502** with the dedicated ITU channel may be an optical channel different in wavelength from sensing channels **2102** and may share the same optical fiber **804/808** where the dedicated communication channel may be coupled to OADM and TX/RX transceiver units proximate to information handling system **146**.

[0098] As discussed above, in FIGS. **21-25**, SPOA **2100** may each have a power supply **2224**. As to be discussed below, power supply **2224** may be a battery or a plurality of batteries, thermo-electric power generation, a remote optically pumped amplification, or remote optically pumped with subsea Optical to Electrical (OE) power conversion.

[0099] For power supply **2224**, a battery or at least one battery within a plurality of batteries may comprise, lithium, alkaline, carbon zinc, gel, nickel oxyhydroxide lead, nickel cadmium, zinc, mercury, calcium, nickel metal hybrid, silver oxide, galvanic, magnesium, and/or the like. In examples, the battery may be a sacrificial anode battery that may comprise magnesium, aluminum, or zinc. A sacrificial anode battery may generate electrical current and power via scavenging electron flow (e.g., electrical current) generated from a cathodic protection system with sacrificial anodes, such as metallic zinc alloys. In examples, cathodic protection system may be designed to last upwards of 50-years before sacrificial anode may be replaced. In most cases underwater well structures may have several tons of anodic mass protection with amperes of potential current flow. Low power remote optical amplification, such as SPOA **2100** (e.g., referring to FIGS. **21-25**), and control systems may utilize only small average DC currents to operate (e.g., 100 mA to 1A) depending on voltage potential created across a shunt resistor within the amplifier housing. Methods to scavenge or generate anode depletion currents may be created by piggybacking to existing subsea rig/tree/distribution center cathodic protection anode infrastructure circuit, via parallel direct connection to sacrificial anode, to provide voltage potential across a series resistor mounted within the metal housing of the amplifier with seawater electrical return from the said metal amplifier housing itself. In other examples, a standalone sacrificial anode to the metal housing of optical flying lead **142** (e.g., referring to FIG. **5**) that may house amplification and control circuits, as this would allow it to be replaceable via ROV (remotely operated vehicle), which may prevent interruption to customer infrastructure. Additionally, a design option where consumable parts may be replaced using an ROV.

[0100] Power supply **2224** may still be a thermo-electric power generator. In examples, a thermo-electric power generator may comprise a housing to collect energy and power optical amplification. During operations, thermo-electric power generator may be permanently deployed as a wrap-around housing disposed in static pipe **114** and/or flexible pipe **118** (e.g., referring to FIG. **1**) during manufacturing/deployment. In a temporary deployment, housing may be a clamp that may be added to any pipe with heat differential, include means for insulation removal. This may be used to add power on demand for both surface and subsea infrastructure where there is a temperature difference. For example, the thermo-electric power generator may access the pipe where the pipe temperature is close to the flowing fluid moving through the pipe where the fluid may have been produced in a reservoir where the temperature is significantly higher than the surrounding ocean water temperature. May producing reservoirs may have temperature above 100 degrees Celsius whereas subsea temperatures may be around 4 degrees Celsius at deep depths thus generating a substantial temperature difference. The surface and subsea infrastructure such as static pipe **114** and/or flexible pipe **118** that is warmer than the water around it may therefore be used to generate a substantial temperature difference across thermo-electric power generator. Using the temperature difference, electricity may be created by placing one thermally conductive heat pad at the hot side of the system and a second thermally conductive heat pad on the cool side of the system where electricity is then generated using e.g. a Peltier element. The power generated by thermo-electric power generators is highly dependent on the fluid temperature and may therefore be lower than desired at start-up conditions, which may impact operation of FOS system **126**.

[0101] In other examples, power supply **224** may be an from the use of optical-electrical conversion of surface pumped optical light locally in the SPOA **2100** (e.g., referring to FIG. **21**) where the electrical power is used to power a 980 nm or 1480 nm single mode pump laser used with the Erbium doped fiber **22100** (e.g., referring to FIG. **22**) in SPOA **2100**.

[0102] The methods and systems describe above are an improvement over current technology where extended reach sensing is enabled by the use of remote power generation, a flexible amplification system that may be optimized for any sensing application, fail safe operation of the SPOAs where a power failure defaults the optical signal path to a low loss signal path away from any Erbium fiber with high signal absorption, the ability to add amplification on any point where there are ROV mate flying leads. The SPOA may be housed in a flying lead in some configurations and deployed and replaced on demand.

[0103] The systems and methods for a fiber optic sensing system discussed above, implemented within a subsea environment may include any of the various features of the systems and methods disclosed herein, including one or more of the following statements. Moreover, the systems and methods for a fiber optic sensing system discussed above implemented within an onshore environment may include any of the various features of the systems and methods disclosed herein, including one or more of the following statements.

[0104] Statement 1: A self-powered optical amplifiers (SPOA) may comprise a proximal tap connected to a fiber optic cable, a proximal wavelength division multiplexer (WDM) optically connected to the proximal tap, and an Erbium doped fiber optically connecting the proximal WDM to a distal WDM. The SPOA may further comprise a distal tap optically connected to the distal WDM, a pump optically connected to the distal WDM, and a controller optically connected to the proximal tap, the distal tap, and further electrically connected to the pump and a power supply.

[0105] Statement 2: The SPOA of statement 1, wherein the SPOA is disposed on a down going fiber.

[0106] Statement 3: The SPOA of statement 2, further comprising a plurality of SPOAs disposed on the down going fiber.

[0107] Statement 4: The SPOA of statement 3, wherein each of the plurality of SPOAs are separated by at least 20 kilometers.

[0108] Statement 5: The SPOA of any previous statements 1 or 2, wherein the SPOA is disposed on an upgoing fiber.

[0109] Statement 6: The SPOA of statement 5, further comprising a plurality of SPOAs disposed on the upgoing fiber.

[0110] Statement 7: The SPOA of statement 6, wherein each of the plurality of SPOAs are separated by at least 20 kilometers.

[0111] Statement 8: The SPOA of any previous statements 1, 2, or 5, wherein the SPOA is disposed on a downhole sensing fiber.

[0112] Statement 9: The SPOA of any previous statements 1, 2, 5, or 8, further comprising an Amplified Stimulated Emission (ASE) filter disposed within the proximal tap.

[0113] Statement 10: The SPOA of any previous statements 1, 2, 5, 8, or 9, further comprising an Amplified Stimulated Emission (ASE) filter disposed within the distal tap.

[0114] Statement 11: The SPOA of any previous statements 1, 2, 5, or 8-10, wherein the pump is a 980 nm pump or 1480 nm pump.

[0115] Statement 12: The SPOA of any previous statements 1, 2, 5, or 8-11, further comprising a 2×2 switch optically connecting the distal tap and the proximal tap.

[0116] Statement 13: The SPOA of statement 12, wherein the 2×2 switch allows for an interrogation signal or backscatter signal to bypass the SPOA.

[0117] Statement 14: The SPOA of any previous statements 1, 2, 5, or 8-12, further comprising a second pump.

[0118] Statement 15: The SPOA of statement 14, further comprising a multiplexer 1×2 optically connects the pump and the second pump to the distal WDM.

[0119] Statement 16: The SPOA of any previous statements 1, 2, 5, 8-12, or 14, wherein the power supply is a thermo-electric power generator.

[0120] Statement 17: The SPOA of any previous statements 1, 2, 5, 8-12, 14, or 16, further comprising an Optical Add Drop Multiplexer (OADM) that is in communication with the controller.

[0121] Statement 18: The SPOA of any previous statements 1, 2, 5, 8-12, 14, or 16-18, wherein the power supply is one or more batteries.

[0122] Statement 19: The SPOA of statement 18, wherein one or more batteries may comprise lithium, alkaline, carbon zinc, gel, nickel oxyhydroxide lead, nickel cadmium, zinc, mercury, calcium, nickel metal hybrid, silver oxide, galvanic, or magnesium.

[0123] Statement 20: The SPOA of statements 18, wherein the power supply is a sacrificial anode that may comprise magnesium, aluminum, or zinc.

[0124] Although the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations may be made herein without departing from the spirit and scope of the disclosure as defined by the appended claims. The preceding description provides various examples of the systems and methods of use disclosed herein which may contain different method steps and alternative combinations of components. It should be understood that, although individual examples may be discussed herein, the present disclosure covers all combinations of the disclosed examples, including, without limitation, the different component combinations, method step combinations, and properties of the system. It should be understood that the compositions and methods are described in terms of "comprising," "containing," or "including" various components or steps, the compositions and methods can also "consist essentially of" or "consist of" the various components and steps. Moreover, the indefinite articles "a" or "an," as used in the claims, are defined herein to mean one or more than one of the element that it introduces.

[0125] For the sake of brevity, only certain ranges are explicitly disclosed herein. However, ranges from any lower limit may be combined with any upper limit to recite a range not explicitly recited, as well as ranges from any lower limit may be combined with any other lower limit to recite a range not explicitly recited, in the same way, ranges from any upper limit may be combined with any other upper limit to recite a range not explicitly recited. Additionally, whenever a numerical range with a lower limit and an upper limit is disclosed, any number and any included range falling within the range are specifically disclosed. In particular, every range of values (of the form, "from about a to about b," or, equivalently, "from approximately a to b," or, equivalently, "from approximately a-b") disclosed herein is to be understood to set forth every number and range encompassed within the broader range of values even if not explicitly recited. Thus, every point or individual value may serve as its own lower or upper limit combined with any other point or individual value or any other lower or upper limit, to recite a range not explicitly recited.

[0126] Therefore, the present examples are well adapted to attain the ends and advantages mentioned as well as those that are inherent therein. The particular examples disclosed above are illustrative only and may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. Although individual examples are discussed, the disclosure covers all combinations of all of the examples. Furthermore, no limitations are intended to the details of construction or design herein shown, other than as described in the claims below. Also, the terms in the claims have their plain, ordinary meaning unless otherwise explicitly and clearly defined by the patentee. It is therefore evident that the particular illustrative examples disclosed above may be altered or modified and all such variations are considered within the scope and spirit of those examples. If there is any conflict in the usages of a word or term in this specification and one or more patent(s) or other documents that may be incorporated herein by reference, the definitions that are consistent with this specification should be adopted.

What is claimed is:

1. A self-powered optical amplifier (SPOA) comprising:

a proximal tap connected to a fiber optic cable;

a proximal wavelength division multiplexer (WDM) optically connected to the proximal tap;

an Erbium doped fiber optically connecting the proximal WDM to a distal WDM;

a distal tap optically connected to the distal WDM;
a pump optically connected to the distal WDM; and
a controller optically connected to the proximal tap, the distal tap, and further electrically connected to and a power supply.

2. The SPOA of claim **1**, wherein the SPOA is disposed on a down going fiber.

3. The SPOA of claim **2**, further comprising a plurality of SPOAs disposed on the down going fiber.

4. The SPOA of claim **3**, wherein each of the plurality of SPOAs are separated by at least 20 kilometers.

5. The SPOA of claim **1**, wherein the SPOA is disposed on an upgoing fiber.

6. The SPOA of claim **5**, further comprising a plurality of SPOAs disposed on the upgoing fiber.

7. The SPOA of claim **6**, wherein each of the plurality of SPOAs are separated by at least 20 kilometers.

8. The SPOA of claim **1**, wherein the SPOA is disposed on a downhole sensing fiber.

9. The SPOA of claim **1**, further comprising an Amplified Stimulated Emission (ASE) filter disposed within the proximal tap.

10. The SPOA of claim **1**, further comprising an Amplified Stimulated Emission (ASE) filter disposed within the distal tap.

11. The SPOA of claim **1**, wherein the pump is a 980 nm pump or 1480 nm pump.

12. The SPOA of claim **1**, further comprising a 2×2 switch optically connecting the distal tap and the proximal tap.

13. The SPOA of claim **12**, wherein the 2×2 switch allows for an interrogation signal or backscatter signal to bypass the SPOA.

14. The SPOA of claim **1**, further comprising a second pump.

15. The SPOA of claim **14**, further comprising a multiplexer optically connects the pump and the second pump to the distal WDM.

16. The SPOA of claim **1**, wherein the power supply is a thermo-electric power generator.

17. The SPOA of claim **1**, further comprising an Optical Add Drop Multiplexer (OADM) that is in communication with the controller.

18. The SPOA of claim **1**, wherein the power supply is one or more batteries.

19. The SPOA of claim **18**, wherein the one or more batteries may comprise lithium, alkaline, carbon zinc, gel, nickel oxyhydroxide lead, nickel cadmium, zinc, mercury, calcium, nickel metal hybrid, silver oxide, galvanic, or magnesium.

20. The SPOA of claim **18**, wherein the power supply is a sacrificial anode that may comprise magnesium, aluminum, or zinc.

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